Forest simulation models

Inputs, PnET, Biomass Succession, HBAD, stands and generative vegetation — calibration and results

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Inputs

Every data source, what it provides, and which model uses it.

Model inputs: what each source provides, and who uses it

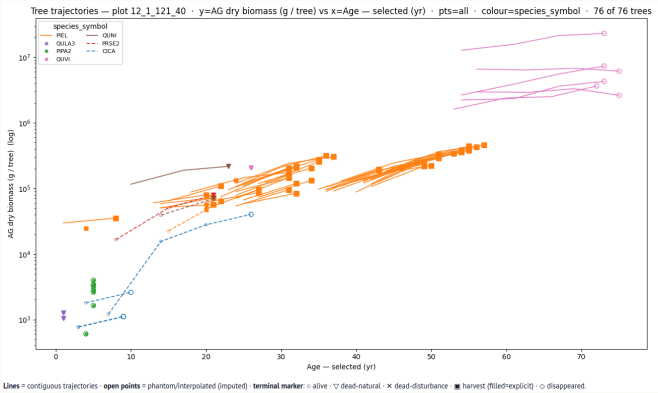
input	provides	BS	PnET	HBAD v1, v2	HBAD3P	viewer
FIA (FIADB)	remeasured plots: trees (DIA, HT, CR, status, TPA), conditions, treatment and disturbance codes	•	•	•	•	•
curated cohorts	per plot visit: (species, imputed age) cohorts with AGB; disturbance drop per cohort	•	•	•		•
FIA-derived traits	longevity (3 classes), shade tolerance (crown class), PROB_ESTAB (canopy cover)	•	•	•	•	
PRISM by EPA L4	monthly Tmin, Tmax, precipitation; normals; baseline forecasts		•		•	•
CO_2 (NOAA, RCMIP)	historical and SSP-scenario atmospheric CO_2		•			•
SOLUS soils	sand, clay, SOC, depth → porosity, field capacity, wilting point, AWC (Saxton–Rawls)		•		•	•
EPA ecoregions L3 / L4	ecoregion cells for parameters; climate aggregation units	•	•	•	•	•
FIA volume equations	cubic-foot and board-foot volume from cohort QMD and height					•

Mapping inputs: what each raster provides, and who uses it

input	provides	stands	gen. veg	viewer
TreeMap 2022	30 m FIA plot imputation: BA, height, QMD, cover, TPA, softwood share; plot id	•	◦	•
Hansen GFC v1.12	tree cover 2000, forest loss year 2001–2024	•	•	
LANDFIRE	EVT type and life form; disturbance class and severity 2000–2025; EVC cover, TPI (held-out scores)	•	•	
AlphaEarth 2022, 2025	64-d annual satellite embeddings (10 m, resampled to 30 m)	•	•	
SOLUS soils	soil PCs (stands), soil covariates of footprints (gen. veg)	•	•	
TIGER, FL parcels	roads, water, parcel lines as stand barriers	•		
FIA public coordinates	fuzzed plot locations (≈1.6 km) for candidate footprints		•	•

• model input or calibration data ◦ comparison only (generative vegetation is TreeMap-independent)

FIA: remeasured trees become cohort trajectories

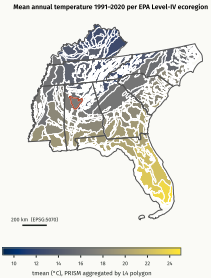


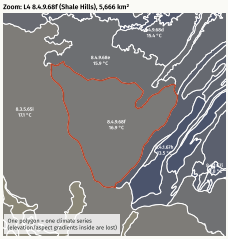
- species resolved, tree ages imputed, fates flagged (mortality, harvest, disturbance)
- cohort = (species, age) on a plot; AGB summed from its trees
- calibration data: GA+FL 8,985 plots, ≈46 species groups; Region 4 ≈30k plots

Data: FIADB TREE / COND / PLOT remeasurements → `curated_trees`, `curated_cohorts_landis_stdorg`. Example: plot 40, Suwannee Co. FL (old-growth live oak with a young slash-pine plantation); figure from the thesis.

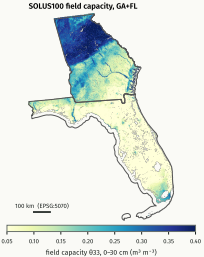
Site drivers: climate and soil (PnET, HBAD3P, viewer)

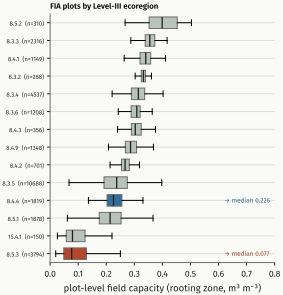
mean temperature by EPA Level-4 ecoregion (PRISM)





field capacity from SOLUS (Saxton–Rawls)

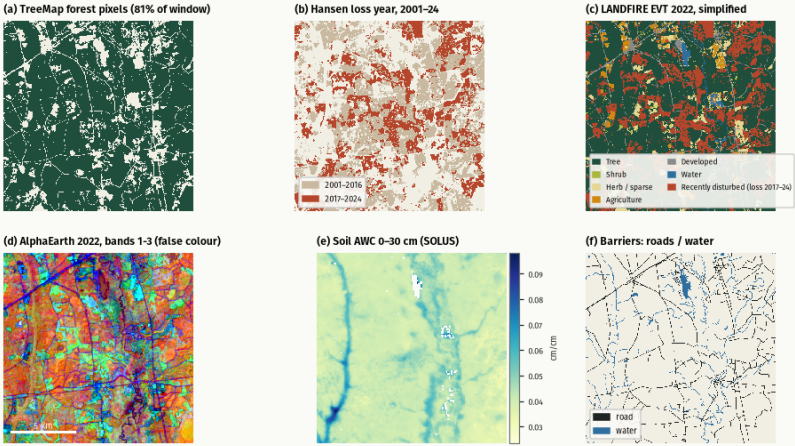




Data: PRISM monthly 1981–2024 aggregated to EPA L4 polygons; SOLUS 0–30 cm texture and SOC at the plot (public) coordinates; Region 4.

Remote-sensing layers (stand delineation, generative vegetation)

Pilot window, south Georgia (15 × 15 km, 30 m grid, EPSG:5070): inputs for microstand segmentation — pipeline not yet run



Data: TreeMap 2022, Hansen GFC v1.12, LANDFIRE EVT 2022, AlphaEarth 2025, SOLUS, TIGER; all resampled to the TreeMap 30 m grid (EPSG:5070).

Climate model: from PRISM to the simulated future

PRISM monthly 4 km, 1895–2025: Tmin, Tmax, Tmean, Tdmean, precip., VPD min/max

→ spatial mean per EPA L4 polygon (967 CONUS; 99 in Region 4)

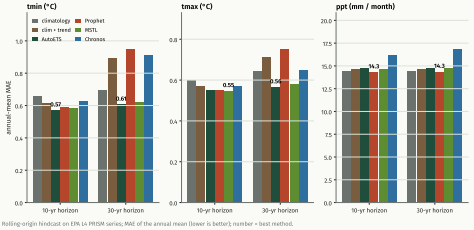
→ **baseline** 2025–2150: AutoETS for Tmin / Tmax (no trend); 1995–2024 climatology for precipitation, VPD

→ **scenario**: user changes (add / multiply, step or ramp, months); Tmax ≥ Tmin, precip. ≥ 0

→ **PnET**: monthly Tmin, Tmax, precip.; derived PAR, VPD, day length

→ **HBAD3P**: W = AET/PET (actual / potential evapotranspiration)

Biomass Succession, HBAD v1: no climate input

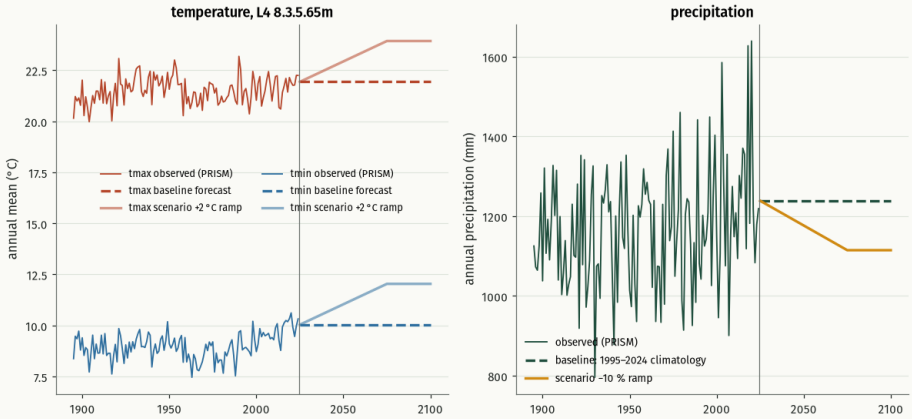


Rolling-origin hindcast on EPA L4 PRISM series; MAE of the annual mean (lower is better); number = best method.

- method chosen by a rolling-origin backtest (20 Region-4 L4s, origins 1964–2014): AutoETS best for Tmin / Tmax at 10 and 30 yr (annual-mean error 0.57–0.61 °C); trends and Prophet worse at 30 yr
- precipitation is unpredictable: every method within ≈3 % ⇒ climatology
- the baseline has no warming trend and no interannual variability: warming / drying enter as scenarios

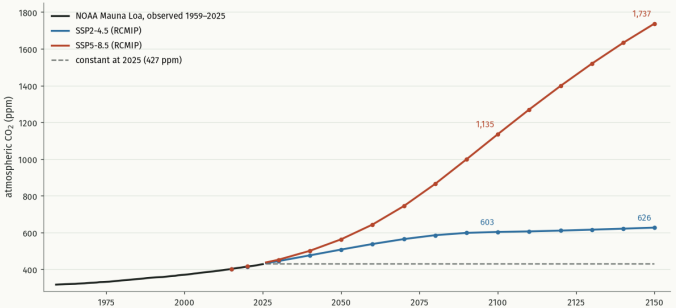
Data: `pan_climate.duckdb`; backtest `scripts/bench_main.csv`; scenario rules `julia/ClimateScenario.jl`.

Climate: observed, baseline and a scenario for one ecoregion



Data: L4 8.3.5.65m (Southeastern Plains), annual means of the monthly series in `pan_climate.duckdb`: PRISM 1895–2024, baseline forecast 2025–2100; example scenario = Tmean +2 °C and precipitation −10 % ramped 2025 → 2075, then held (ClimateScenario `ramp`). Scenarios are what Forest Explorer users set.

CO_2: NOAA observations and SSP pathways



- observed years: NOAA Mauna Loa annual means (calibration, hindcasts)
- future: SSP2-4.5 or SSP5-8.5 global-mean concentrations (RCMIP v5.1.0, Meinshausen et al. 2020), decadal values interpolated; or constant / linear
- in PnET: Amax $\times \frac{C_i - 40}{C_i + 80} \cdot \frac{430}{310}$, $C_i = C_a(0.875 - 0.075\,\mathrm{FolN})$; water-use efficiency rises with $C_a - C_i$

Data: `landis_region4/data_r4/co2_annual.csv` (NOAA MLO 1959–2025); `pan_climate/README.md` (RCMIP values); Pan `PnET/plugin.jl` (`sublayer!`: `delamax`, `jco2`).

Preliminary: the thesis

Thesis: Biomass Succession calibrated with multi-objective CMA-ES on the Southern Coastal Plain of Florida (EPA L3 8.5.3).

MO-CMA-ES with two simulation schemes (Sim A, Sim B)

FIA cohort trajectories (species, age, AGB) per plot visit

Sim A — deterministic: cohorts injected when first observed; no reproduction; fits S, D, ANPP$_{\max}$, $B_{\max}$

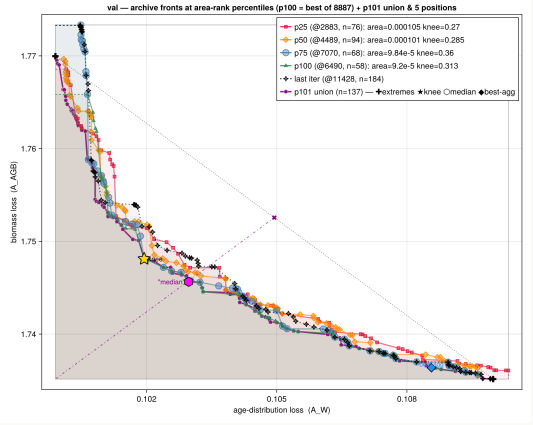
Sim B — stochastic: cohorts establish by seeding; replicate runs; adds PROB_ESTAB, PROB_MORT

two objectives: AGB discrepancy + Wasserstein-1 distance of the age-class biomass distribution

(μ+μ) MO-CMA-ES from Igel's (1+1)-CMA-ES: non-dominated rank, crowding; IPOP restarts

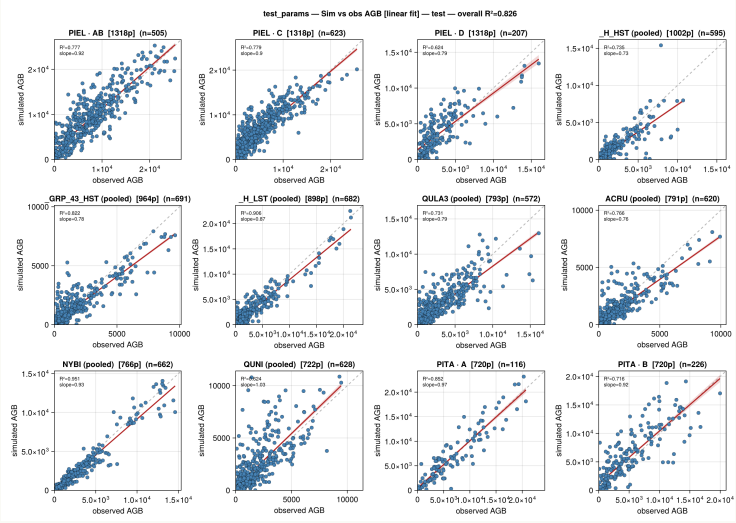
select on the validation front; TOST on held-out test

validation Pareto fronts, FL 8.5.3 (candidate sweep)



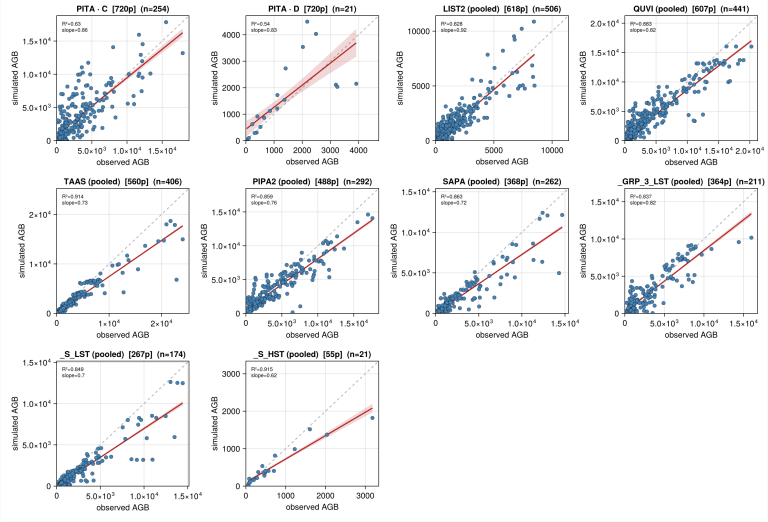
Data: FIA GA+FL curated cohorts, EPA L3 8.5.3 (Southern Coastal Plain, FL); run `fl853_igelmo_4cell_shade_n1000` (pines split into site-class cells A–D).

FL 8.5.3 held-out test: simulated vs observed AGB (1/2)



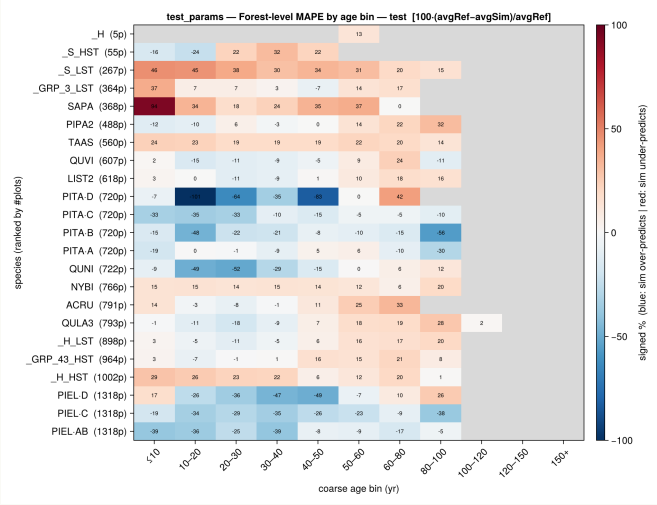
Data: candidate 6, held-out test split; cohort AGB (g/m^2) per (plot, visit, species group); overall R^2 = 0.826.

FL 8.5.3 held-out test: simulated vs observed AGB (2/2)



Data: as before; panels ordered by number of plots (in brackets), *n* = test pairs.

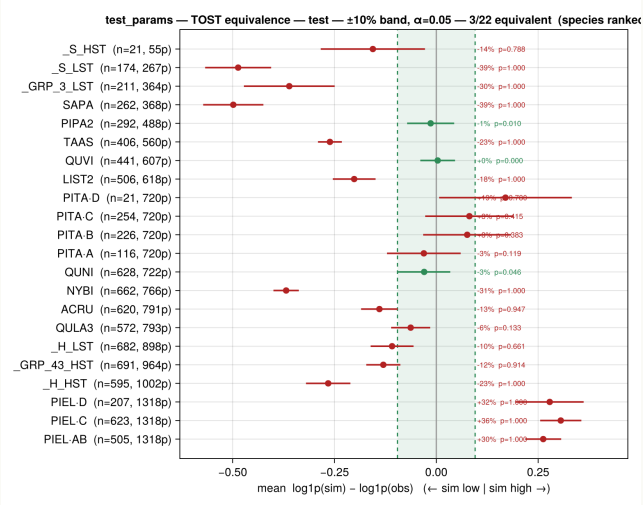
FL 8.5.3 held-out test: forest-level comparison by age class



- signed % error of the mean biomass per species and 10–20-yr age class (red: under-predicted)
- young loblolly (PITA·D) and slash pine over-predicted; other low-tolerance softwoods (_S_LST), cabbage palm (SAPA), pond cypress (TAAS) under-predicted
- most hardwood groups within ±20 %

Data: candidate 6, test split; $100\,(\overline{\text{obs}} - \overline{\text{sim}})/\overline{\text{obs}}$ per cell; grey = no data.

FL 8.5.3 held-out test: equivalence (TOST, ±10 %)

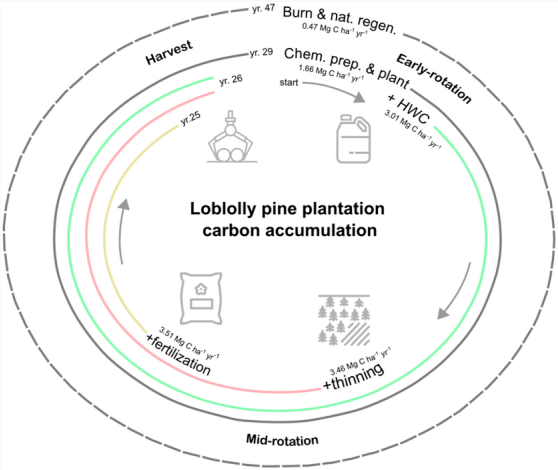


- 3 of 22 groups equivalent at ±10 %: water oak (QUNI), live oak (QUVI), longleaf pine (PIPA2)
- at ±15 % also laurel oak (QULA3) and loblolly class A (PITA·A)
- the productive pines fail through their persistent biomass spread: slash pine (PIEL) over-, other low-tolerance softwoods (_S_LST) and cabbage palm (SAPA) under-predicted

Data: candidate 6, held-out test split; per group mean $\log(1+\text{sim}) - \log(1+\text{obs})$ with 90 % CI, α 0.05; n = pairs, p = plots. Thesis figure *fig:fl853_cand6_tost*.

Why pines miss: one species, very different growth curves

- Under management the same species grows on very different site curves: productive plantations vs gradual natural growth.
- One (S, D, $\mathrm{ANPP}_{\max}$, $B_{\max}$) set per species cannot span both ⇒ a finer stratification (separate growth parameters by management / site) is needed before judging Biomass Succession inflexible.
- Loblolly pine (PITA): treatment shortens the time to commercial maturity from 47 yr (natural) to 25 yr (fertilized).



Data: figure from Shephard, Narine, Peng & Maggard (2022), *Climate Smart Forestry in the Southern United States*, Forests 13(9):1460 (as in the thesis deck): loblolly pine plantation carbon accumulation by silvicultural regime.

FL 8.5.3: fitted parameters (validation candidates, mean ± sd, n=137)

Species	D	Lng	S	ANPP$_{\max}$	$B_{\max}$
ACRU	18.5 ± 3.1	150	0.198 ± 0.181	450 ± 53	15635 ± 1846
LIST2	15.6 ± 4.0	400	0.064 ± 0.073	407 ± 62	14126 ± 2116
NYBI	17.1 ± 5.1	275	0.807 ± 0.127	632 ± 144	20082 ± 4343
PIEL AB	8.9 ± 4.2	275	0.196 ± 0.201	809 ± 30	27753 ± 425
PIEL C	8.9 ± 4.2	275	0.196 ± 0.201	662 ± 13	22924 ± 366
PIEL D	8.9 ± 4.2	275	0.196 ± 0.201	404 ± 12	13825 ± 183
PIPA2	19.3 ± 5.9	275	0.471 ± 0.381	468 ± 113	15086 ± 2536
PITA A	17.0 ± 7.0	275	0.096 ± 0.091	894 ± 20	30781 ± 186
PITA B	17.0 ± 7.0	275	0.096 ± 0.091	896 ± 44	30897 ± 1166
PITA C	17.0 ± 7.0	275	0.096 ± 0.091	748 ± 69	24807 ± 726
PITA D	17.0 ± 7.0	275	0.096 ± 0.091	504 ± 60	12950 ± 1231

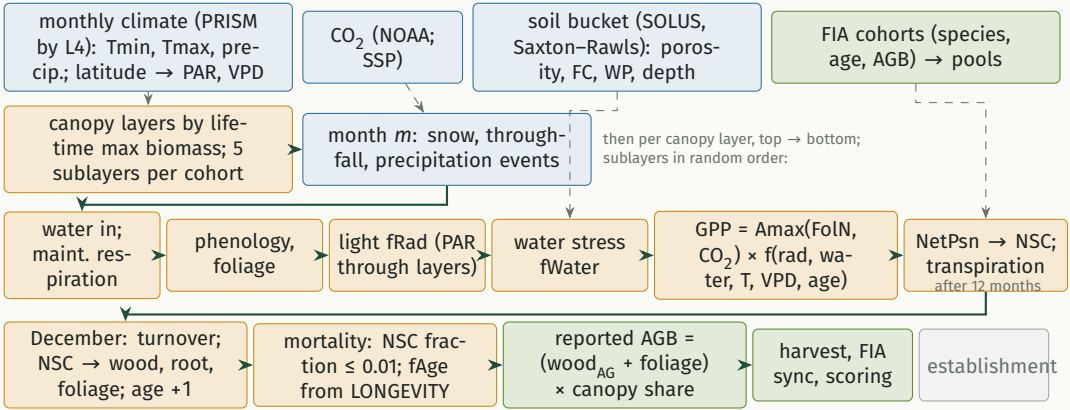
Species	D	Lng	S	ANPP$_{\max}$	$B_{\max}$
QULA3	20.3 ± 2.4	150	0.208 ± 0.123	599 ± 37	19499 ± 1963
QUNI	19.2 ± 5.8	275	0.107 ± 0.144	520 ± 94	16279 ± 1650
QUVI	19.3 ± 4.6	400	0.587 ± 0.185	879 ± 190	27612 ± 6423
SAPA	17.5 ± 5.1	275	0.921 ± 0.057	980 ± 109	31330 ± 2630
TAAS	16.6 ± 5.4	400	0.905 ± 0.063	925 ± 165	28254 ± 3719
_GRP_3_LST	20.1 ± 5.1	275	0.46 ± 0.115	514 ± 118	17578 ± 4159
_GRP_43_HST	18.3 ± 4.6	275	0.311 ± 0.272	477 ± 122	15913 ± 3869
_H_HST	17.4 ± 4.8	275	0.131 ± 0.173	439 ± 89	14742 ± 3087
_H_LST	17.6 ± 4.9	275	0.229 ± 0.209	430 ± 20	14780 ± 583
_S_HST	18.0 ± 5.5	275	0.764 ± 0.361	907 ± 399	22685 ± 7081
_S_LST	20.5 ± 4.6	275	0.875 ± 0.13	841 ± 176	25141 ± 2429

Data: p101 validation candidates of the FL 8.5.3 run. D mortality curve, S growth curve, Lng longevity (yr); ANPP$_{\max}$ and $B_{\max}$ in g m^{-2}. PITA A–D / PIEL AB–D are site-productivity cells. Table from the thesis deck.

PnET-Succession

Physiology: monthly photosynthesis from foliar N, light, water, temperature, VPD and CO_2 on the Biomass Succession cohort frame.

PnET-Succession in Pan: one simulated year



Pan `src/plugins/PnET/plugin.jl` (`site_year!`, `sublayer!`); `docs/pnet_spec.md` §1.4, §7 (from the LANDIS-II PnET-Succession v6 code)

PnET: variables and how they are computed

variable	definition / equation (Pan code)	from which input
pools	foliage, wood+root (TotalBiomass), NSC per cohort; solved so that reported AGB = FIA AGB	FIA cohort AGB, age
T_{ave}, T_{day}	$(T_{min} + T_{max})/2$; $(T_{max} + T_{ave})/2$	PRISM monthly, by L4
VPD (kPa)	$e_s(T_{day}) - e_s(T_{min})$ (Magnus)	PRISM
PAR	Bristow–Campbell transmissivity (from $T_{max} - T_{min}$) × FAO-56 extraterrestrial radiation × 0.45 × 4.57 / day length	PRISM, latitude
fRad	$1 - \exp(-\text{PAR}_{layer} \ln 2 / \text{HalfSat})$, $\text{PAR}_{layer} = \text{PAR}\, e^{-k\,\text{LAI}_{above}}$	PAR, canopy LAI
FC, WP, porosity	soil water at 33 and 1500 kPa, saturation (Saxton–Rawls)	SOLUS sand, clay, SOC
fWater	Feddes response to the soil pressure head (H1–H4) of the current bucket water	soil, precipitation, transpiration
Amax	AmaxA + AmaxB·FolN (CO_2 modifier)	FolN (fitted), CO_2
fAge	$\max(0, 1 - (a/L)^{\text{PsnAgeRed}})$	age, LONGEVITY
GPP, NetPsn	Amax · fRad · fWater · fT(T_{day}) · f_{VPD} · fAge; NetPsn = GPP – foliar respiration → NSC	all of the above
NSC fraction	NSC / (FActiveBiom · (wood + foliage) · C fraction); cohort dies at ≤ 0.01	state
reported AGB	$(\text{wood}_{AG}$ + foliage) × canopy-layer share	state

Pan `src/plugins/PnET/climate.jl` (VPD, PAR), `plugin.jl` (`sublayer!` §5.5–5.8), `site_attrs.jl` (soils); `docs/pnet_spec.md` §1–8

PnET vs Biomass Succession in operation, and its parameters

	Biomass Succ.	PnET
time step	annual	monthly (12 per year)
growth	$ANPP_{max}$ × competition × S	GPP from FolN, light, water, T, VPD, CO_2 – respiration
competition	share of site biomass, B_{max}	light by canopy layer, one soil-water bucket
mortality	age curve D; growth-related	carbon starvation, turnover
establishment	PROB_ESTAB × light	off in Pan
cohort	(species, age, AGB)	+ foliage, wood/root, NSC
shared	cohort frame, LONGEVITY, FIA injection, harvest, loss	

Fitted (GA+FL, Sim A)

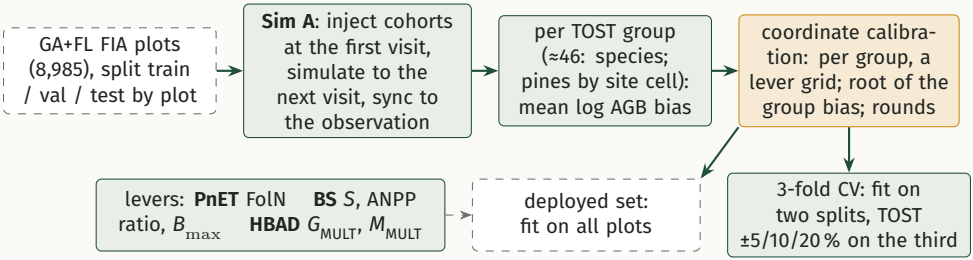
- **FolN** (foliar N %, eco × species group) — the productivity lever: Amax = AmaxA + AmaxB·FolN
- search space of the CMA-ES runs: + MaxFracFol, FrActWd, SLWmax, TOwood, PsnAgeRed (species), RD_MULT (eco): 282 dims

Fixed

- LONGEVITY (as BS); species → 4 plant functional types (22 physiology fields: AmaxA/B, HalfSat, k, PsnT, H1–H4, DVPD, TOfol, …)
- 21 generic constants; site: climate, CO_2, soils

Pan `src/plugins/PnET/params.jl`; `docs/pnet_spec.md` §2; `docs/cmaes_gdigel.md:13`

Calibration procedure (Sim A; same for PnET, Biomass Succession, HBAD v1)

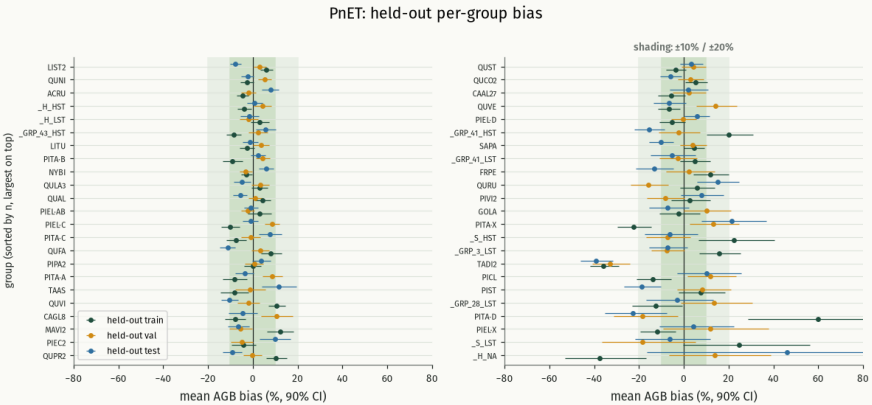


- matched pairs = (plot, visit, group), real records only; disturbed cohorts excluded
- TOST: equivalence of the group's mean log bias within ±δ (α 0.05)

Caveat: root-finding the group bias optimises the statistic TOST tests ⇒ only held-out folds are informative. The non-TOST redo (multi-objective, next part) is in progress.

Pan `tools/calib_coord.jl`, `tools/tost_common.jl`, `tools/gafl_cv.sh`; `docs/gafl_calibration_log.md`, `docs/gafl_pnet_calibration.md`

PnET calibration: held-out bias per species group



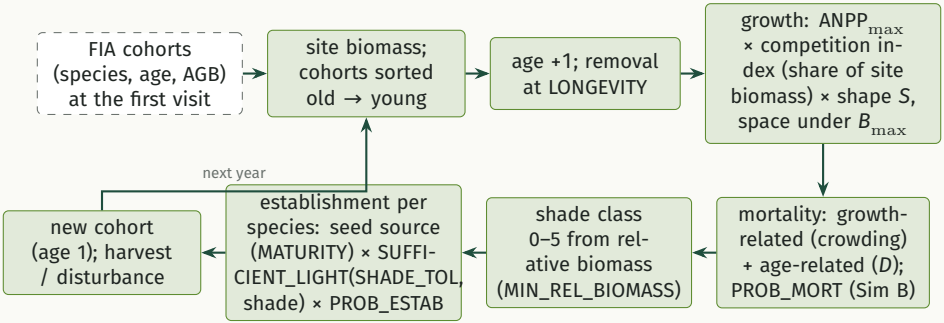
held-out groups passing ±5/10/20 %: **3 / 16 / 33** of 46 (mean of folds); median |bias| 6.1 %; bald cypress (TADI2) cannot be fitted: its PFT, not FolN, limits it

Data: `results/pnet_gafl/cvP_*_groups.csv`: each fold held out from a FolN calibration on the other two splits; 90 % CI. **TOST-targeted calibration** (held-out folds shown)

Biomass Succession

The LANDIS-II core model: biomass cohorts, ANPP / B_{max} growth, age- and growth-related mortality, establishment.

Biomass Succession in Pan: one simulated year



Sim A (growth / decay): FIA cohorts injected at each visit, establishment off, PROB_MORT = 0 — deterministic.

Sim B (free run): establishment on, PROB_MORT stochastic, replicate runs; no sync after the first visit.

Pan `src/plugins/BiomassSuccession/plugin.jl` (`succession_step!`, `reproduction_check_step!`); LANDIS-II Biomass Succession v7 guide; Scheller & Mladenoff (2004)

Biomass Succession: variables and how they are computed

variable	definition / equation (Pan code)	from which input
cohort, age a	(subplot, species, birth date = visit − imputed age); age propagated in integer steps per subplot	FIA TREE, age anchors (TOTAGE, BHAGE, STDAGE)
B_i **(g/m²)**	cohort AGB = $\sum_{\text{trees}}$ DRYBIO_AG × TPA_UNADJ × 453.592/4046.86	FIA TREE
eco cell	EPA L3 ecoregion × site-class tier (pines also by site cell A–D)	COND (SITECLCD), EPA L3
comp$_i$	$\max(1, B_i^{0.95}) \,/\, \sum_j \max(1, B_j^{0.95})$	state
$B_{\text{POT},i}$, $B_{\text{AP},i}$	$B_{\max} - B_{\text{site}} + B_i$; $B_{\text{AP}} = B_i / B_{\text{POT}}$	state, $B_{\max}$
ANPP$_i$	$\text{ANPP}_{\max}\ \text{comp}_i\ B_{\text{AP}}^{S}\, e^{1-B_{\text{AP}}^{S}}$, $\text{ANPP}_{\max} = B_{\max}/\text{ratio}$	state, S, $B_{\max}$, ratio
M_{growth}, M_{age}	$\text{ANPP}_{\max}\ \text{comp}_i\ \frac{2B_{\text{AP}}}{1+B_{\text{AP}}}$; $B_i\, e^{D(a/L-1)}$ (L = LONGEVITY)	state, D, L
ΔB_i	$\text{ANPP}_i - M_{\text{growth}} - M_{\text{age}}$ (LANDIS v7 order); removal at $a \geq L$	—
shade class	number of MIN_REL_BIOMASS thresholds exceeded by $B_{\text{site}} / \max B_{\max}$	state
establishment	mature cohort present × SUFFICIENT_LIGHT[shade, SHADE_TOL] × PROB_ESTAB	traits from FIA crown class, canopy cover

Pan `BiomassSuccession/plugin.jl` (`succession_step!`); `FIA_DATA_PREP/cohort_tracking.py`, `cohorts_landis.py`

Biomass Succession parameters

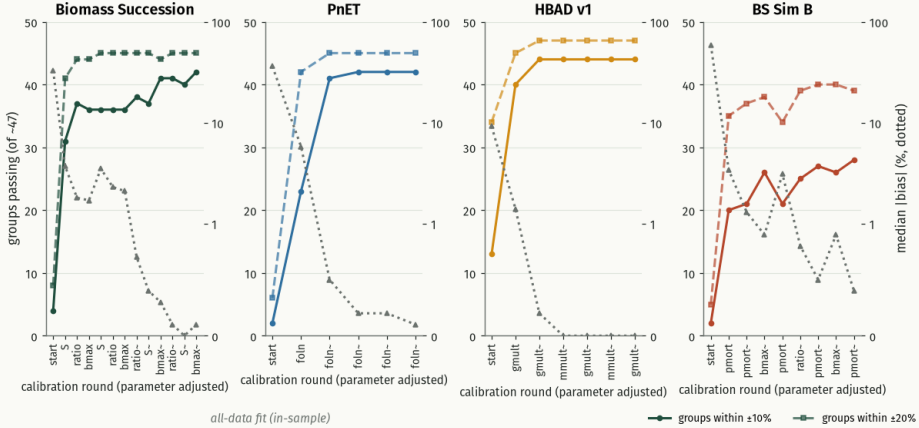
	parameters	from
fitted, Sim A	S (growth shape), $B_{\max}$, ANPP ratio $B_{\max}/\mathrm{ANPP}_{\max} \in [20, 35]$ (eco × species group); D (species)	coordinate calibration (deployed); CMA-ES / gdigel: 224 dims
fitted, Sim B	PROB_MORT (+ $B_{\max}$, ratio refined)	free-run calibration
fixed from data	LONGEVITY: 150 / 275 / 400 yr (PLANTS classes); SHADE_TOL 1–5 (FIA crown class); PROB_ESTAB (canopy-cover window)	FIA + SONA / FEIS / PLANTS
defaults	SUFFICIENT_LIGHT (data-derived taper), MIN_REL_BIOMASS, MATURITY	LANDIS guide, data

ecoregion cells: EPA L3 × site class; pines (PITA, PIEL) split by site-productivity cell (A–D).

thesis values for FL 8.5.3: preliminary part (parameter table).

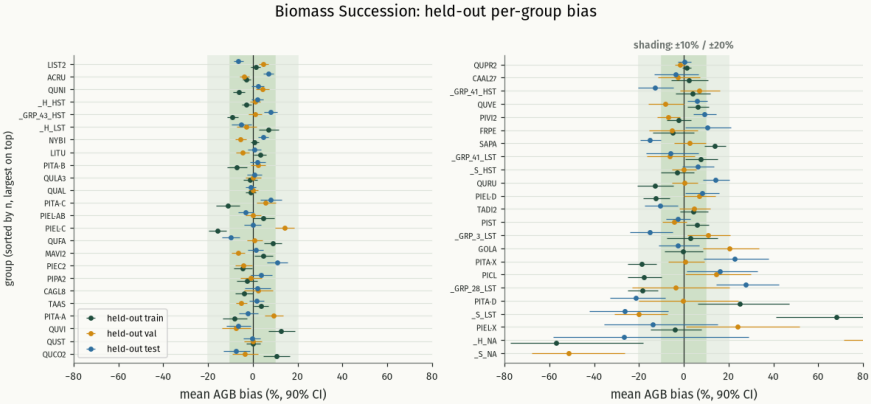
Pan `src/plugins/BiomassSuccession/params.jl`; `tools/calib_levers.jl`; `docs/gafl_calibration_log.md` D1–D15; `docs/cmaes_gdigel.md:13`

Calibration rounds: all-data fits of the four set-ups



Data: `results/{gafl, pnet_gafl, hbad_gafl}/final_calib_rounds.csv`, `results/gafl_simB/final_calib_rounds_3lever.csv`: groups passing and median |bias| after each lever round, all GA+FL plots (in-sample). **TOST-targeted calibration** (held-out folds shown)

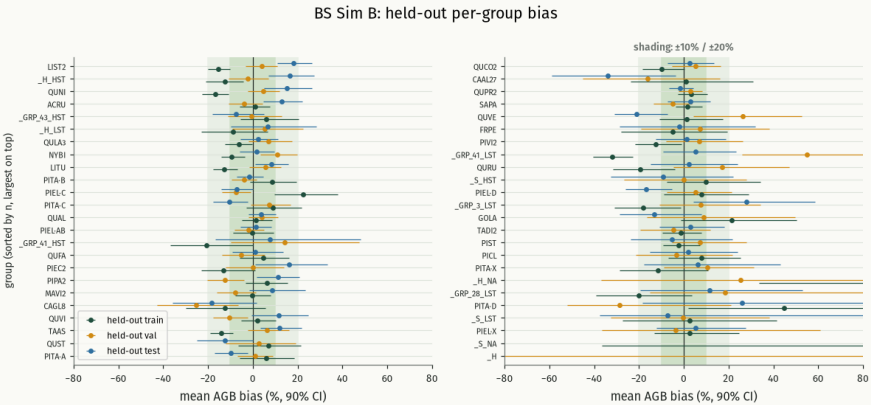
Biomass Succession (Sim A): held-out bias per species group



held-out groups passing ±5/10/20 %: **8 / 20 / 36** of 46 (mean of folds); median |bias| 5.1 %; small catch-all groups (_S_NA, _H_NA) fail

Data: `results/gafl/cv_heldout_{train,val,test}_groups.csv`; levers S, ratio, $B_{\max}$. **TOST-targeted calibration** (held-out folds shown)

Biomass Succession (Sim B, free run): held-out bias per species group



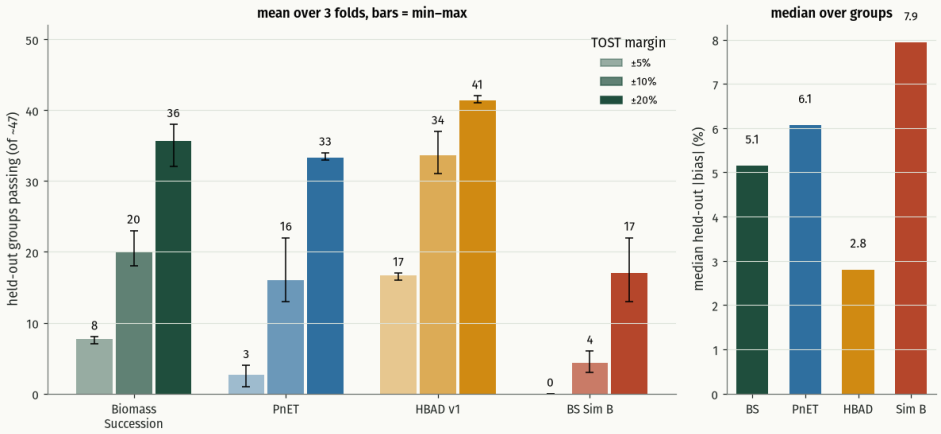
held-out groups passing ±5/10/20 %: **0 / 4 / 17** of 48; median |bias| 7.9 % — a free stochastic run cannot predict which cohorts die

Data: `results/gafl_simB/cvB_*_groups.csv`: plot-cluster bootstrap of $\log \Sigma\text{sim}/\Sigma\text{obs}$ per group; levers PROB_MORT, $B_{\max}$, ratio. **TOST-targeted calibration** (held-out folds shown)

Calibration results compared

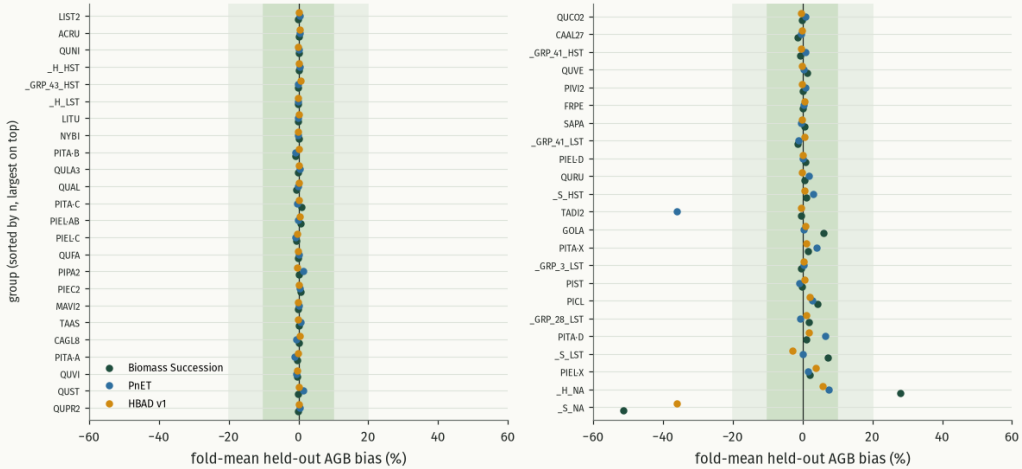
Same data, same groups, same splits: the engines compared on held-out plots — and the non-TOST recalibration.

GA+FL: held-out equivalence of the three engines



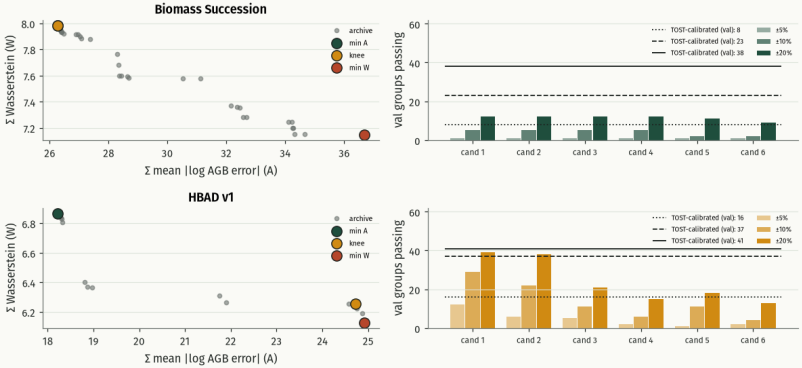
Data: the three CV folds of each engine (`results/*/cv*_groups.csv`); groups ≈46 (Sim A), 48 (Sim B). HBAD v1 is best held-out; PnET equals BS in-sample but generalises slightly worse. **TOST-targeted calibration** (held-out folds shown)

Per species group: where the engines differ



Data: fold-mean held-out bias per group (same files); bands ±10 / ±20 %. Large groups agree within a few %; the engines diverge on small groups: catch-alls (_S_NA, _H_NA) and bald cypress (TADI2, PnET).

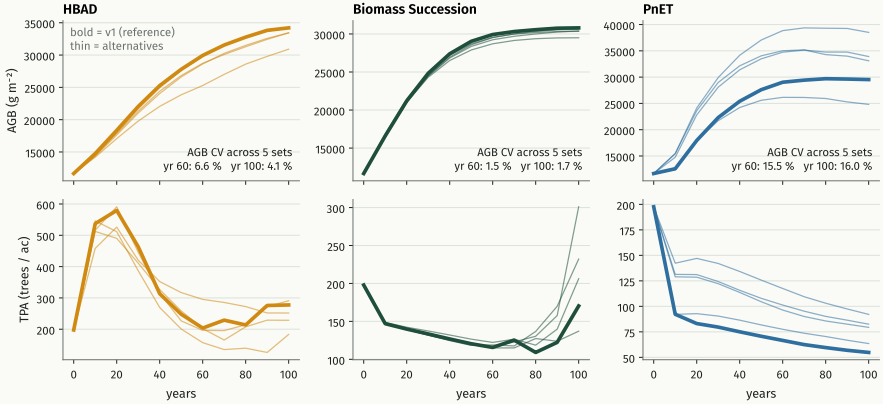
Redo without TOST: multi-objective gdigel (accuracy + age structure)



- objectives per group: a_g = mean | log AGB error|, w_g = Wasserstein age-structure distance; picks (min *A*, knee, min *W*, refined) scored on *val* only
- **not yet better**: BS val 1/5/12 vs 8/23/38; HBAD 12/29/39 vs 16/37/41 (TOST-targeted). Sim A/B variants (6, BS): none passes all criteria

Data: `runs_local/mogd_{bs,hbad}_legacyW/mo_archive.csv`, `mogd_*_val.log`; references from the val CV folds; `docs/mo_gdigel.md`, `docs/simAB_options_results.md`.

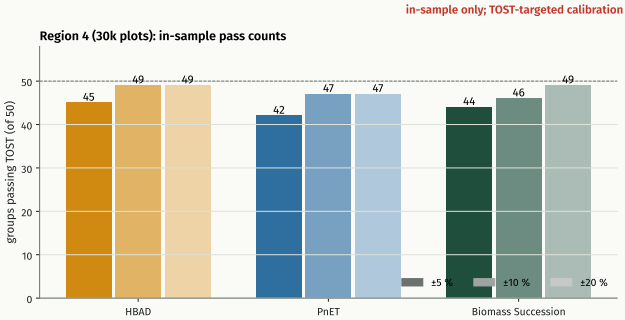
Equifinality: equally good parameter sets, different futures



100-yr grow-only projections on 60 GA/FL plots. PnET alternatives are independent gdigel solutions (not the anchored exploration).

Data: gdigel solution ensembles (`results/explore_{biomass,pnet,hbad}`), 100-yr projections from the same start; spread at year 60: BS CV ≤1.7 %, HBAD 6.6 %, PnET ≈15 %.

Region 4: all three engines on ≈30k plots



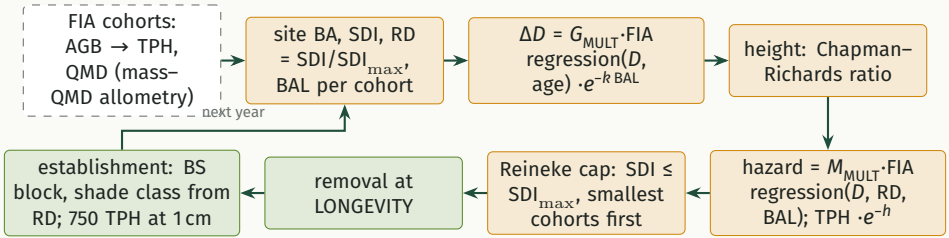
- 8 states, 14 EPA L3 ecoregions, 50 species groups
- same procedure, all plots: in-sample only
- the viewer's default parameter sets

Data: `results/r4/{bs,pnet,hbad}_calib_log.csv`; `docs/campaign_2026-10-14.md`. **TOST-targeted calibration** (held-out folds shown)

HBAD

Height, Basal Area, Density: cohorts carry trees/ha, diameter and height — v1 empirical, v2 process, HBAD3P process and trait-bound.

HBAD v1 in Pan: one simulated year



Fitted: G_{MULT}, M_{MULT} (eco × species), SDI$_{max}$ (species). **Fixed**: 28 FIA regression coefficients per species. No climate, CO_2 or soil drivers.

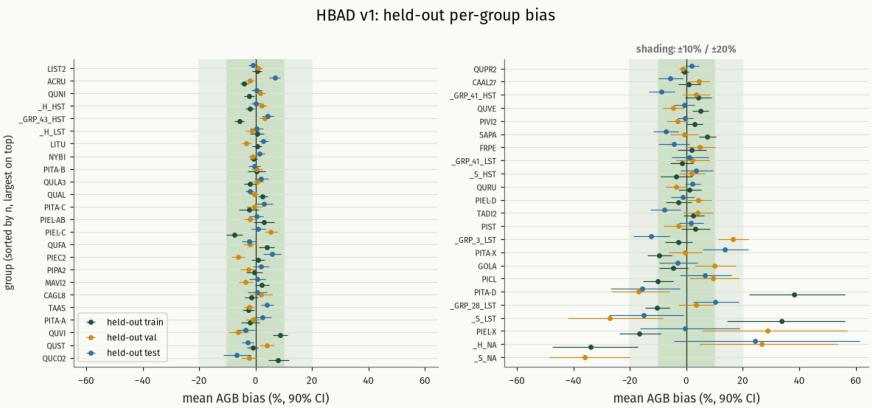
Pan `src/plugins/HBAD/plugin.jl`; `docs/hbad_plugin.md` §1–5; equations: backup

HBAD v1: variables and how they are computed

variable	definition / equation (Pan code)	from which input
TPH (trees/ha)	from cohort (age, AGB): $\exp$(quadratic in $\ln$ AGB, $\ln$ age), ≥ one macroplot tree	FIA cohort AGB, age; fitted translation
QMD D (cm)	mass per tree m = AGB/TPH → $D = 2.54\, e^{(\ln m - a)/b}$ (mass–QMD allometry)	FIA trees (allometry fit)
height H (m)	$1.3 + h_a(1 - e^{-h_b D})^{h_c}$ (Chapman–Richards)	FIA HT (prior fit)
BA, SDI	$\frac{\pi}{4}(D/100)^2$ TPH; $\sum$ TPH $(D/25.4)^{1.605}$ (Reineke)	state
RD	SDI / $\mathrm{SDI}_{\max}$ (site value from species $\mathrm{SDI}_{\max}$, BA-weighted)	$\mathrm{SDI}_{\max}$ (fitted)
BAL	BA of cohorts with larger D	state
ΔD	$G_{\mathrm{MULT}} \exp(g_0 + g_1 \ln \frac{D}{20} + g_2 \frac{D}{20} + g_3 \ln \frac{A}{30} - g_k \frac{\mathrm{BAL}}{10})$	FIA remeasured trees (28 coeff./species)
hazard h	$M_{\mathrm{MULT}} \exp(m(D, \mathrm{RD}, \mathrm{BAL}))$; TPH ← TPH e^{-h}	FIA remeasured trees
AGB (reported)	TPH · $m(D)$ (same allometry)	state
shade class	from RD vs MIN_REL_RD (establishment block of Biomass Succession)	state

Pan `docs/hbad_plugin.md` §1–4, `src/plugins/HBAD/plugin.jl`, `tools/hbad_translate_lib.jl`

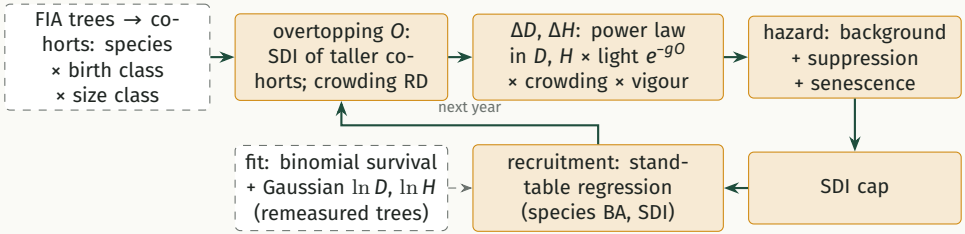
HBAD v1 calibration: held-out bias per species group



held-out groups passing ±5/10/20 %: **17 / 34 / 41** of 46 (mean of folds); median |bias| 2.8 % — best of the three engines; growth–mortality split not identified (M_{MULT} ×11 across equal fits)

Data: `results/hbad_gafl/cvH_*_groups.csv`; levers G_{MULT}, M_{MULT}. **TOST-targeted calibration** (held-out folds shown)

HBAD v2: process cohorts with light competition by height



No biomass state: N (trees/ha), D, H, age, vigour quantile. ≈14 parameters per species key, ≈175–210 in all. GA+FL val NLL −272,655.

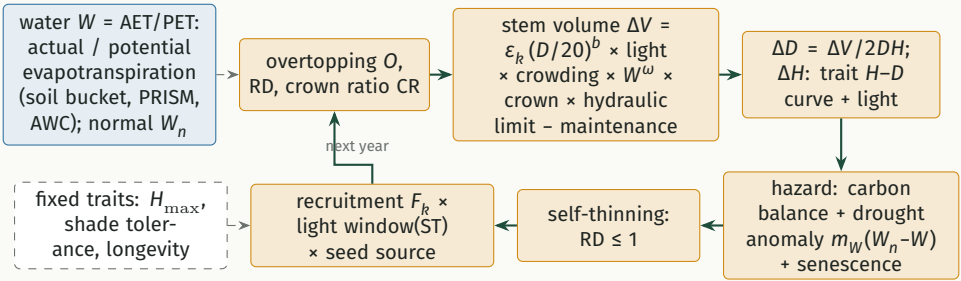
`landis_hbad2` `docs/hbad_v2.md`, `src/plugins/HBAD2/HBAD2.jl`; equations: backup

HBAD v2: variables and how they are computed

variable	definition / equation (code)	from which input
cohort	species key × 10-yr birth class × size class, with vigour quantile z; N trees/ha, D QMD, H mean height, A age	remeasured FIA trees (undisturbed intervals)
SDI, RD	$\sum N(D/25.4)^{1.605}$; $\mathrm{SDI}/\mathrm{SDI}_{\max}$	state
overtopping O	SDI of taller cohorts (half of equal-height ones and of itself) / $\mathrm{SDI}_{\max}$	state
ΔD	$r_D(D/20)^{c_D} e^{-D/D_{\max}} e^{-g_{OD}O} e^{g_{RD}\mathrm{RD}} e^{\sigma_v z}$	fitted per key
ΔH	$r_H(H/15)^{c_H} e^{-H/H_{\max}} e^{-g_{OH}O+g_{RH}\mathrm{RD}} e^{\sigma_v z}$	fitted per key
hazard h	background $e^{m_{bg}+m_D \ln \frac{D}{20}+m_{D2}\frac{D}{20}}$ + suppression (light deficit) + senescence $e^{m_A(A/L-1)}$	fitted per key, LONGEVITY
recruits R	$185.2\exp(\rho_0 + \rho_S \ln(1 + \mathrm{BA}_{sp,D\geq 12.7}) + \rho_L \mathrm{SDI}/600)$ trees/ha/yr at 2.54 cm, 3 m	FIA saplings (stand-table regression)
fit	binomial survival + Gaussian $\ln D$, $\ln H$ of survivors over each remeasurement interval	FIA TREE remeasurements

`landis_hbad2` `docs/hbad_v2.md`, `src/plugins/HBAD2/HBAD2.jl`

HBAD3P: process and trait-bound growth, climate through water



Fitted: growth efficiency ε_k per key + globals (33 coordinates; v2: 175) by CMA-ES; fecundity F_k and the light window in a separate recruitment fit. GA+FL val NLL −273,469 vs v2 −272,655 on the same data; the deployed pipeline fit (forested-area densities, 42 coordinates) −272,208 on its own data. In Pan: plugin `hbad3p` (bit-identical to the reference); Forest Explorer engine, climate scenarios act through W each year.

`landis_hbad3` `docs/hbad3_design.md` §4, `results/hbad3/fit3p_P3_GAFL/log.txt`; Pan `src/plugins/HBAD3P`; equations: backup

HBAD3P: variables (1/2: state)

variable	definition / equation (code)	from which input
site	one FIA subplot (4 per plot)	FIA PLOT / SUBP
cohort	(species key, birth decade $\lfloor (\text{year} - \text{age})/10 \rfloor$, D class [5, 12.7, 25, 40] cm) × 5 vigour nodes z (Gauss–Hermite)	`curated_trees` (live, aged)
N **(trees/ha)**	$4 \sum$ TPA_UNADJ·2.47105 × node weight $/ a$; a = $\max(f, 0.25)$, f = forested share of the subplot	TREE, SUBP_COND × COND
D, H, A, **CR**	TPA-weighted quadratic mean DIA·2.54; mean HT·0.3048 (≥1.37); mean age; mean CR/100	TREE (DIA, HT, CR), imputed age
SDI, RD	$\sum N(D/25.4)^{1.605}$; $\text{SDI}/\text{SDI}_{\max}$	state
overtopping O	$\left(\sum_{H_j > H_i} \text{SDI}_j + \frac{1}{2} \sum_{H_j = H_i} \text{SDI}_j + \frac{1}{2}\text{SDI}_i\right)/\text{SDI}_{\max}$	state

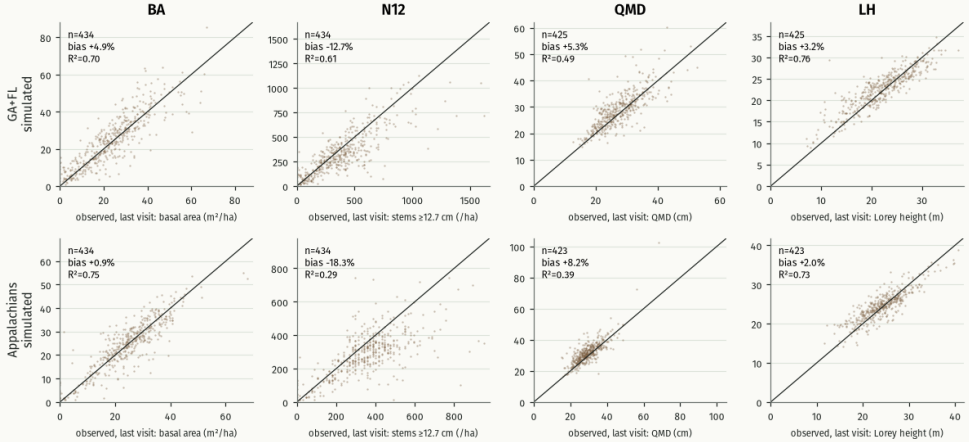
Pan `tree_cohorts3p.jl`, `plugins/HBAD3P`; `landis_hbad3` `docs/hbad3_design.md` §4, `climate_join.py`

HBAD3P: variables (2/2: water, growth, mortality, recruits)

variable	definition / equation (code)	from which input
PET (mm/month)	*potential evapotranspiration*: water a well-watered surface would lose. Thornthwaite: $16\,(10T/I)^a \cdot \frac{N}{12} \cdot \frac{\text{days}}{30}$, I = annual heat index, N = day length	monthly mean T (PRISM / scenario), latitude
AET (mm/month)	*actual evapotranspiration*: what the soil supplies. Bucket of capacity AWC: S += P; AET = $\min(\text{PET}, S)$; S = $\min(\text{AWC}, S - \text{AET})$	precipitation P, AWC (SOLUS)
water W, W_n	April–October ΣAET/ΣPET (1 = no water limit), 5-yr trailing mean in scenarios; W_n = 1991–2020 mean	the two rows above
ΔV, ΔD	$\varepsilon_k (D/20)^b \times$ light(O)× crowding(RD) $\times W^\omega \times$ vigour(z)× crown(CR) × hydraulic($H/H_{\max}$) − $\rho D^2 H$; $\Delta D = \Delta V / 2DH$	state, traits, W
hazard h	$f_M[m_0/L + m_s e^{\sigma_{ST}(\text{ST}-3)} e^{-\beta g_m} (D/20)^\zeta] + e^{m_A(A/L-1)}$; drought: $f_M \propto e^{m_W(W_n - W)/0.1}$	state, traits, W
recruits R_k	$F_k \exp(-\lambda_0 e^{\lambda_1(\text{ST}-3)}\text{RD}) \times$(seed source ? 1 : ε_0) $\times f/a$	state, shade tolerance

Pan `tree_cohorts3p.jl`, `plugins/HBAD3P`; `landis_hbad3` `docs/hbad3_design.md` §4, `climate_join.py`

HBAD3P free runs on held-out plots



Data: `results/hbad3/fit3p_P3_{GAFL,APP}/freerun_val.csv`: 434 val plots per region, simulated from the first visit to the last (8–30 yr, median 20) without sync; bias = Σsim/Σobs − 1. Stems ≥12.7 cm too low (−13 / −18 %): in-growth still under-predicted.

The three HBAD generations side by side

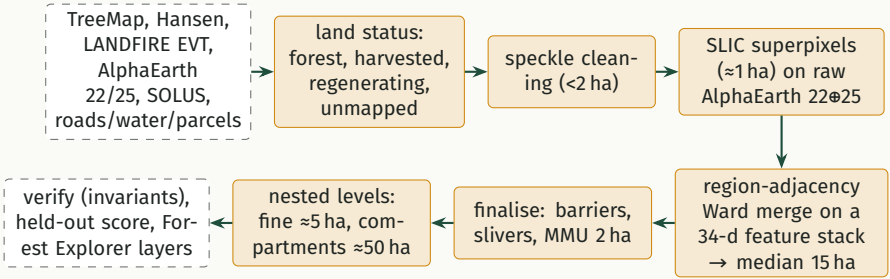
	HBAD v1	HBAD v2	HBAD3P
growth	FIA ΔD regression $\times G_{\text{mult}}$, BAL	power law in D, H × light (O), crowding (RD), vigour	stem-volume balance: light, crowding, water W^{ω}, crown, hydraulic limit
height	Chapman–Richards ratio	own growth equation	trait H–D curve + light rule
mortality / thinning	regression hazard $\times M_{\text{mult}}$ (RD, BAL); Reineke cap	background + suppression + senescence; SDI cap	carbon-balance hazard + drought anomaly + senescence; RD ≤ 1
recruitment	Biomass-Succession block	stand-table regression	fecundity × light window × seed source
climate	none	none	W (PRISM + AWC), drought anomaly
parameters	G, M per eco×spp, $\text{SDI}_{\max}$	≈ 175–210	≈ 30–33 per region
fit (GA+FL val NLL)	—	−272,655	−273,469

fit: Sim A likelihood on remeasured trees (binomial survival + Gaussian ln D, ln H), CMA-ES, same GA+FL data for v2 and HBAD3P. HBAD v1 is calibrated on cohort AGB instead (held-out TOST, previous part). HBAD3P free runs: previous slide.

Stand delineation

Stands from satellite embeddings and land-status rasters: where the simulation units on the map come from.

Stand delineation: the pipeline



Superpixels and merges never cross barriers or land status. One config, content-hashed stage cache; `stands_pipeline.py` for any region.

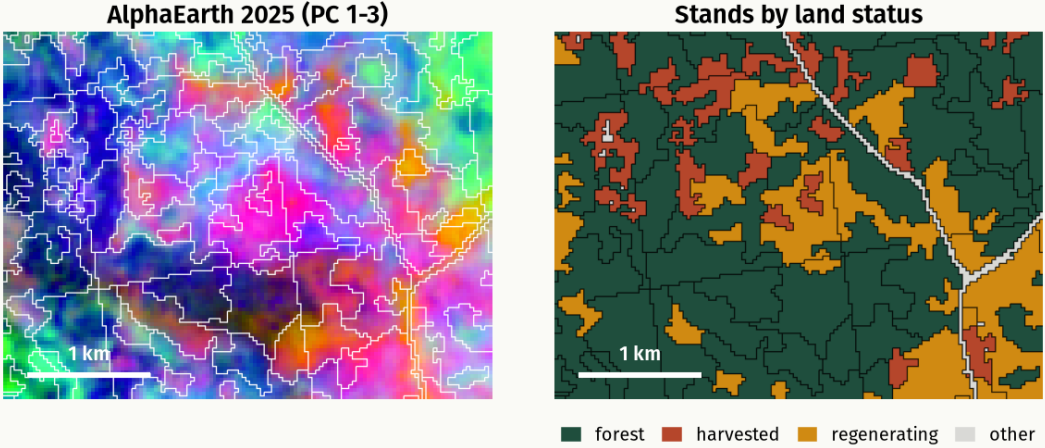
`standmap` `docs/STANDS_PIPELINE.md`, `config/pilot.yml`; formulation: backup

Stand delineation: variables and how they are computed

variable	definition / equation (code)	from which input
land status	1 forest (TreeMap pixel); 2 harvested (loss ≥2017 on cover$_{2000}$ ≥25 %); 3 regenerating (older loss, disturbed / plantation EVT, or shrub/herb on former forest); 4 forest unmapped (EVT tree); 9 converted	TreeMap, Hansen loss year + cover 2000, LANDFIRE EVT
AlphaEarth $\mathbf{a}$	64-d embedding, de-quantised $\mathrm{sign}(q)(q/127.5)^2$, 10 m → 30 m	AlphaEarth 2022, 2025
superpixel	SLIC on $\mathbf{a}_{22} \oplus \mathbf{a}_{25}$ (128-d), $n = n_{\mathrm{px}}/11$ (≈1 ha), compactness 0.15, per status and tile	AlphaEarth
feature y **(34-d)**	AEF 2025: 16 PCs (weight 1.0); AEF 2022 on the 2025 basis: 8 (0.4); change $1-\cos(\mathbf{a}_{25}, \mathbf{a}_{22})$ (0.5); TreeMap: 6 z-scores (0.6); soils: 3 PCs (0.3)	AlphaEarth, TreeMap (ln BA, height, ln QMD, cover, ln TPA, softwood share), SOLUS
merge cost c_{ab}	Ward $\frac{n_a n_b}{n_a+n_b}\lVert\mu_a-\mu_b\rVert^2 \times s_{ab}$, s_{ab} favours long shared boundaries; threshold to a 15 ha median	superpixel means of y
barriers	1-px lines no superpixel or stand crosses	TIGER roads, water; FL parcels
stand attributes	status, area, loss year, TreeMap means, main TreeMap plot, soils	all of the above
score (share)	$(R^2_{\mathrm{stands}} - R^2_{\mathrm{blocks}})/(1 - R^2_{\mathrm{blocks}})$ on held-out layers	Hansen canopy 2000, LANDFIRE EVC, slope, TPI

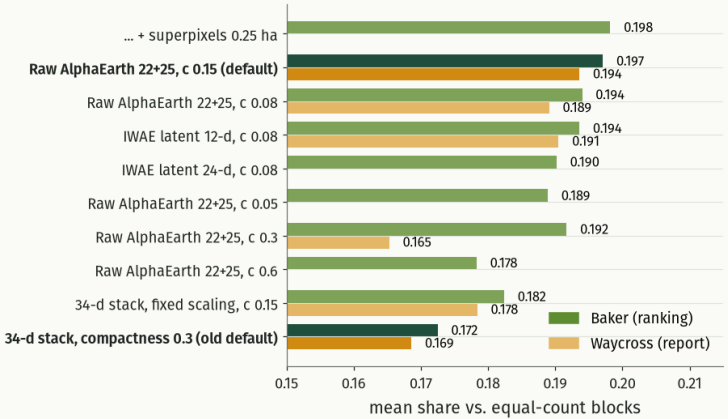
`standmap` `scripts/10_forest_land.py`, `features.py`, `21_superpixels.py`, `22_merge.py`, `23_finalize.py`, `stands_metrics.py`; `config/pilot.yml`

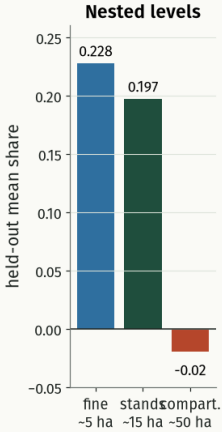
Stands on the map (Baker Co., FL)



Data: default run (`runs/gafl_pilot`): `stands.tif`, land status; AlphaEarth 2025 PCs of this 4×3 km crop; 114 stands.

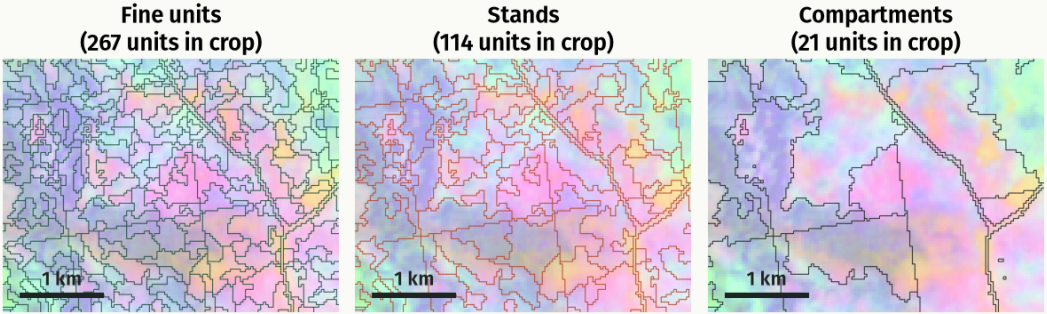
Tuning: which input makes better stands





Data: sweeps `tune_slic`, `tune_embed(2)`, `tune_hier_a` (`results.csv`, `report_window.csv`). Share = $(R^2_{\text{stands}} - R^2_{\text{blocks}})/(1 - R^2_{\text{blocks}})$ on layers the segmentation does not use (canopy 2000, EVC cover, slope, TPI), vs equal-count square blocks; ranked on Baker, reported on Waycross GA.

Nested levels: fine units, stands, compartments

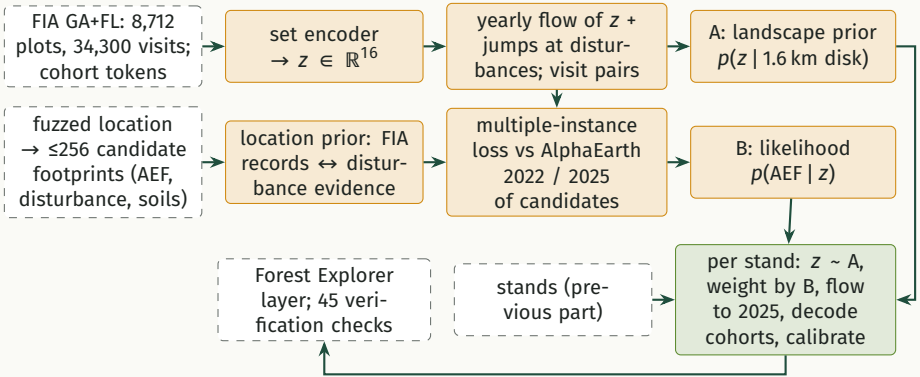


Data: same crop; levels stage of the default run: 98,814 fine units, 39,324 stands, 17,638 compartments (Baker window). Fine units explain the held-out layers best (share 0.23); compartments only as well as blocks.

Generative vegetation

Stand vegetation (cohorts) generated from rasters, learned from FIA plots whose true locations are unknown — independent of TreeMap.

Generative vegetation: model and mapping



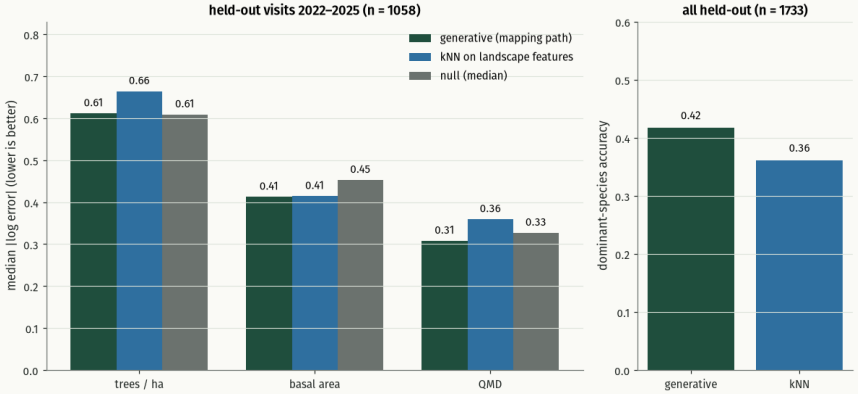
`standmap` `scripts/g02*–g05*`, `docs/GENVEG_PIPELINE.md`; formulation: backup

Generative vegetation: variables and how they are computed

variable	definition / equation (code)	from which input
visit tokens	one per (species, age class): $\ln N$, $\ln$QMD, H, CR per forested area of the visit	FIA trees, conditions, imputed ages
candidates c	90 m footprints of forest land within ≈1.6 km of the public coordinates, ≤256 per plot (stratified, inclusion-weighted)	FIA public coordinates, land status
footprint evidence	AlphaEarth 2022 / 2025; disturbance history 2000–2025 (class, severity, loss year); soils	AlphaEarth, LANDFIRE disturbance, Hansen, SOLUS
location prior π_c	product of likelihood ratios matching FIA treatment / disturbance codes to the footprint's evidence by type and year (LR 4 or 2; ×0.25 for unrecorded stand-replacing events)	FIA TRT / DSTRB codes
latent $z \in \mathbb{R}^{16}$	masked set encoder of the visit tokens	tokens
dynamics	yearly flow $\frac{dz}{dt} = -(\frac{1}{T} + g) \odot z + g \odot A$; jump $z \leftarrow z + J(z, \text{event})$ at the candidate's disturbances	disturbance history
MIL distance d_c	$\lVert h_{22}(z_{2022}) - \mathbf{a}_{22}^c \rVert^2 + \lVert h_{25}(z_{2025}) - \mathbf{a}_{25}^c \rVert^2$, loss $-\log \frac{1}{K} \sum_c e^{-d_c/T}$	AlphaEarth of the candidates
stand vegetation	landscape descriptor X_L (1.6 km disk / 1 km node) → $z \sim p(z \mid X_L)$, weighted by $p(\text{AEF} \mid z)$ of the stand's footprints, flowed to 2025, decoded to cohorts; × calibration factors	stands, rasters, calibration plots

`standmap/scripts`: `g02b`, `g02e`, `g03b`, `g04`, `g05`

Generative vegetation: held-out accuracy



Modest: beats kNN on QMD and the null on BA; no better than the null on trees/ha. Dominant species 0.42 vs kNN 0.36 (majority class 0.22).

Data: `data/proto/gafl_pipe/bayes_eval.json`: held-out FIA visits in spatial blocks (3.2 km buffer), predicted through the stand-mapping path with calibration factors from separate calibration plots.

Backup: formulation

Backup: HBAD v1 — empirical cohort model (FIA regressions + structural competition)

State per cohort: species, age, TPH, QMD D (cm), height H; site: BA = $\frac{\pi}{4}(D/100)^2$ TPH, SDI = $\sum \text{TPH}\,(D/25.4)^{1.605}$, RD = SDI/$\text{SDI}_{\max}$, BAL = BA of larger cohorts.

$$\Delta D = G_{\text{mult}} \exp\!\left(g_0 + g_1 \ln \tfrac{D}{20} + g_2 \tfrac{D}{20} + g_3 \ln \tfrac{A}{30} - g_k \tfrac{\text{BAL}}{10}\right) \in [0, 3]\ \text{cm/yr}$$

$$H' = 1.3 + (H - 1.3)\,\frac{\text{CR}(D')}{\text{CR}(D)}, \qquad \text{CR}(D) = h_a\left(1 - e^{-h_b D}\right)^{h_c}$$

$$h = M_{\text{mult}} \exp\!\left(m_0 + m_1 \ln \tfrac{D}{20} + m_2 \tfrac{D}{20} + m_3 \text{RD}_c + m_4 \tfrac{\text{BAL}}{10} + m_5 \text{RD}_c \ln \tfrac{D}{20}\right) \le 0.5$$

$$\text{TPH} \leftarrow \text{TPH}\, e^{-h}, \qquad \text{RD}_c = \min(\text{RD}, 1.5)$$

- Reineke self-thinning: while SDI > $\text{SDI}_{\max}$, remove trees from the smallest cohorts up; cohorts end at longevity.
- Establishment: Biomass-Succession block (PROB_ESTAB; light class from RD; new cohort 750 TPH, 1 cm).
- **Fitted**: G_{mult}, M_{mult} (ecoregion × species), $\text{SDI}_{\max}$ (species). **Fixed**: the regression coefficients (FIA GA+FL).
- **Inputs**: FIA cohorts (age, AGB → TPH, QMD; AGB via $\ln m = a + b \ln(D/2.54)$); no climate, CO_2 or soil drivers.

landis_fe docs/hbad_plugin.md, src/plugins/HBAD/plugin.jl

Backup: HBAD v1 — where the equations come from

component	form	established form (usual reference)	ours / fitted to FIA
stand density	SDI = $\sum \text{TPH}(D/25.4)^{1.605}$, RD = SDI/$\text{SDI}_{\max}$	Reineke (1933)	$\text{SDI}_{\max}$ per species (calibrated)
self-thinning	remove trees until SDI ≤ $\text{SDI}_{\max}$	Reineke boundary	smallest cohorts first
diameter growth	$\ln \Delta D$ linear in $\ln D$, D, $\ln A$, BAL	log-linear increment with BAL competition, FVS / Prognosis (Wykoff 1990)	covariate set, centring D/20, A/30, BAL/10, cap 3 cm/yr; coefficients FIA
height	$h_a(1 - e^{-h_b D})^{h_c}$; $H' = 1.3 + (H - 1.3)\frac{\text{CR}(D')}{\text{CR}(D)}$	Chapman–Richards (Richards 1959; Chapman 1961)	ratio rule; coefficients FIA
mortality	hazard $h = e^{X\beta}$, survival e^{-h}; X: $\ln D$, D, RD_c, BAL, $\text{RD}_c \ln D$	binomial GLM, complementary log-log link, $\log \Delta t$ offset	covariates, caps $\text{RD}_c \le 1.5$, $h \le 0.5$; coefficients FIA
multipliers	G_{MULT}, M_{MULT} (eco × species)	—	Pan calibration levers
biomass	$\ln m = a + b \ln(D/2.54)$, AGB = TPH·$m$	log-log allometry (form of Jenkins et al. 2003)	coefficients FIA
cohort translation	TPH from (AGB, age): quadratic in $\ln$AGB, $\ln$age	—	form ours; coefficients FIA
establishment	PROB_ESTAB × light class × seed source	LANDIS-II Biomass Succession (Scheller & Mladenoff 2004)	shade class from RD

Established = a published, widely used form; "ours" = a design choice of this project; "FIA" = coefficients estimated from GA+FL FIA remeasurements. References given from memory: check before presenting.

Backup: HBAD v2 — biomass-free process cohorts with light competition by height

State per cohort: N (trees/ha), D, H, age A, vigour quantile z. Overtopping by taller cohorts (relative density units):

$$O_i = \frac{\sum_{H_j > H_i} \mathrm{SDI}_j + \frac{1}{2}\sum_{H_j = H_i} \mathrm{SDI}_j + \frac{1}{2}\mathrm{SDI}_i}{\mathrm{SDI}_{\max}}$$

$$\Delta D = r_D\,(D/20)^{c_D}\, e^{-D/D_{\max}}\, e^{-g_{OD}O}\, e^{g_{RD}\mathrm{RD}}\, e^{\sigma_v z}$$

$$\Delta H = r_H\,(H/15)^{c_H}\, e^{-H/H_{\max}}\, e^{-g_{OH}O + g_{RH}\mathrm{RD}}\, e^{\sigma_v z}$$

$$h = f_M\Big[e^{m_{bg} + m_D \ln\frac{D}{20} + m_{D2}\frac{D}{20}} + m_{\sup}\big(1 - \min(1, e^{-g_{OD}O} e^{\sigma_v z})\big)\big(\tfrac{D}{20}\big)^{m_{SD}}\Big] + e^{m_A(A/L - 1)}$$

$$N \leftarrow N\,e^{-h}$$

$$R = 185.2\,\exp\big(\rho_0 + \rho_S \ln(1 + \mathrm{BA}_{sp,\,D \geq 12.7}) + \rho_L\,\mathrm{SDI}/600\big)\ \text{trees/ha/yr}$$

- **Parameters**: ≈ 14 per species key ($H_{\max}$, r_H, c_H, $D_{\max}$, r_D, c_D, g_{OD}, g_{RD}, g_{OH}, g_{RH}, m_{bg}, m_D, $m_{\sup}$, $\mathrm{SDI}_{\max}$), globals, recruitment: ≈ 175–210 in all. Recruits enter at 2.54 cm, 3 m.
- **Inputs**: remeasured FIA trees (undisturbed intervals), cohorts by species × 10-yr birth class × size class; **fit**: binomial survival + Gaussian $\ln D$, $\ln H$ of survivors.

landis_hbad2 docs/hbad_v2.md, src/plugins/HBAD2/HBAD2.jl

Backup: HBAD v2 — where the equations come from

component	form	established form (usual reference)	ours / fitted to FIA
cohort state	N, D, H, age, vigour z; key × birth decade × size class	size-structured cohort models	classes; 5 Gauss–Hermite vigour nodes
overtopping O	SDI of taller cohorts / $\mathrm{SDI}_{\max}$	asymmetric competition by taller trees: BAL (Wykoff 1990); height-structured light (Purves et al. 2008)	SDI units; ranking by cohort mean height
growth	$r(D/20)^c e^{-D/D_{\max}} e^{-g_O O} e^{g_R \mathrm{RD}} e^{\sigma_v z}$ (same form for ΔH)	potential × modifiers (FVS, 3-PG); power–exponential increment shape (cf. Zeide 1993)	separate light / crowding terms for D and H (FIA finding: crowding shifts growth to height); 14 parameters per key, FIA
light modifier	$e^{-g_O O}$	exponential attenuation (Beer–Lambert; Monsi & Saeki 1953)	SDI as leaf-area proxy
mortality	background $e^{m_{bg}+m_D \ln \frac{D}{20}+m_{D2}\frac{D}{20}}$ + suppression + $e^{m_A(A/L-1)}$	senescence: LANDIS-II age mortality (Scheller & Mladenoff 2004); U-shaped size term	suppression term; coefficients FIA
self-thinning	SDI cap	Reineke (1933)	—
recruitment	$185.2 \exp(\rho_0 + \rho_S \ln(1 + \mathrm{BA}_{sp}) + \rho_L \mathrm{SDI}/600)$	185.2 = FIA microplot expansion (trees/ha per sapling)	regression form; coefficients FIA
estimation	binomial survival + Gaussian $\ln D$, $\ln H$ per interval	maximum likelihood; CMA-ES (Hansen & Ostermeier 2001)	FIA remeasurement intervals

Established = a published, widely used form; "ours" = a design choice of this project; "FIA" = coefficients estimated from GA+FL FIA remeasurements. References given from memory: check before presenting.

Backup: HBAD3P — process and trait-bound growth (stem-volume balance)

State as v2 + crown ratio CR. W = growing-season AET/PET (actual / potential evapotranspiration), W_n = plot normal.

$$\Delta V = \varepsilon_k\,(D/20)^b \cdot 122.98\ \underbrace{\left[f_0 + (1-f_0)e^{-aO}\right]}_{\text{light}}\ e^{g_R \mathrm{RD}}\ \underbrace{W^{\omega}}_{\text{water}}\ e^{\sigma_v z}\left(\frac{\mathrm{CR}}{0.4}\right)^{\varphi}$$

$$\times\ \underbrace{\max\!\left(0,\,1-(H/1.1H_{\max})^{\theta}\right)}_{\text{hydraulic}}\ -\ \underbrace{\rho\,D^2H}_{\text{maintenance}}$$

$$\Delta D = \frac{\Delta V}{2DH}$$

$$H^*(D) = 1.37 + (H_{\max} - 1.37)\left(1 - e^{-sD/(H_{\max}-1.37)}\right)^c$$

$$\Delta H = \left[H^*(D+\Delta D) - H^*(D)\right]\left(1+\eta(1-L_h)\right) + \lambda\left(H^*(D)(1+\mu(1-L_h)) - H\right), \quad L_h = e^{-k_h O}$$

$$\mathrm{CR}^* = \mathrm{cr}_{lo} + (\mathrm{cr}_{hi} - \mathrm{cr}_{lo})\,e^{-k_c O}, \qquad \mathrm{CR} \to \mathrm{CR}^* \text{ at rates } \tau_r \text{ (recession)},\ \tau_e \text{ (expansion)},\ \mathrm{CR}_0 = \text{FIA CR}$$

- Per species **traits fixed** ($H_{\max}$, shade tolerance ST, longevity L); per key only the growth efficiency ε_k and fecundity F_k are fitted. 122.98 is a fixed unit constant of ε.
- Densities per ha of the *forested* subplot; one pipeline `pipeline3p.sh <REGION>` with 37 checks.

landis_hbad3 docs/hbad3_design.md §4, src/plugins/HBAD3/HBAD3P.jl (step_site!)

Backup: HBAD3P — mortality, thinning, recruitment, climate

$$g_m = e^{-a_m O}\, e^{g_R \mathrm{RD}}\, W^{\omega}\, e^{\sigma_v z}\, (\mathrm{CR}/0.4)^{\varphi} \qquad \text{(survival carbon balance)}$$

$$h = f_M\Big[\frac{m_0}{L} + m_s\, e^{\sigma_{ST}(\mathrm{ST}-3)}\, e^{-\beta g_m}\Big(\frac{D}{20}\Big)^{\zeta}\Big] + e^{m_A(A/L-1)}$$

$$f_M = e^{p_M[\text{planted}]}\; e^{m_W(W_n - W)/0.1} \qquad \text{(drought anomaly)}$$

$$N \leftarrow N\, e^{-h} \qquad \text{and, while RD > 1: remove SDI} \propto \mathrm{SDI}_i(1 - v_i)$$

$$R_k = F_k\; \exp\!\big(-\lambda_0 e^{\lambda_1(\mathrm{ST}-3)}\,\mathrm{RD}\big) \times (\text{seed source}\,?\,1 : \varepsilon_0)$$

fitted on sapling ingrowth × probe-recruit survival to the next visit

Climate inputs

- growth: W^{ω} (water limitation)
- mortality: drought anomaly $m_W(W_n - W)$
- CO_2, temperature: not in the model (FIA cannot identify them; planned as literature-fixed scenario factors)

Parameters: ≈ 30–33 per region (vs ≈ 175–210 in v2):

- per key: ε_k, F_k
- growth: b, a, g_R, θ, ω, a_m, ρ, f_0, φ
- height: s, c, η, λ, μ, k_h
- mortality: m_0, m_s, σ_{ST}, β, ζ, m_W
- planted p_D, p_H, p_M; vigour σ_v; crown; recruitment

src/plugins/HBAD3/HBAD3P.jl; docs/hbad3_design.md §4, §7

Backup: HBAD3P — where the equations come from

component	form	established form (usual reference)	ours / fitted to FIA
volume balance	ΔV = production $-\rho D^2 H$; $\Delta D = \Delta V / 2DH$	carbon-balance growth: 3-PG (Landsberg & Waring 1997), PnET (Aber & Federer 1992)	stem-volume form; $\rho \approx 0$ in the fits
size scaling	production $\propto \varepsilon_k (D/20)^b$; leaf area $\propto D^{1.605}$	pipe model (Shinozaki et al. 1964)	b (1.76 GA+FL), ε_k: FIA
light	$f_0 + (1 - f_0) e^{-aO}$	Beer–Lambert attenuation	diffuse floor f_0; SDI as leaf-area proxy
water	W^ω, W = growing-season AET/PET	multiplicative modifiers (3-PG, PnET); PET of Thornthwaite (1948)	monthly bucket on AWC; ω: FIA (1.20 GA+FL, 0 APP)
hydraulic limit	$\max(0, 1 - (H/1.1H_{\max})^\theta)$	hydraulic limitation of tall trees (Ryan & Yoder 1997; Koch et al. 2004)	functional form; θ: FIA
crown	$(\mathrm{CR}/0.4)^\varphi$; $\mathrm{CR} \to \mathrm{cr}_{lo} + (\mathrm{cr}_{hi} - \mathrm{cr}_{lo}) e^{-k_c O}$	crown recession under shade (concept)	form; parameters FIA
height	$H^*(D)$ + shade elongation (η, λ, μ)	Chapman–Richards curve	elongation rule; parameters FIA
mortality	$f_M[\frac{m_0}{L} + m_s e^{\sigma_{ST}(\mathrm{ST}-3)} e^{-\beta g_m} (\frac{D}{20})^\zeta] + e^{m_A(A/L-1)}$	growth-dependent mortality (e.g. Bigler & Bugmann 2003); LANDIS-II senescence	drought anomaly $m_W(W_n - W)$ (m_W 0.46 GA+FL, 0.75 APP), shade-tolerance scaling; FIA
self-thinning	RD ≤ 1, least vigorous first	Reineke (1933)	removal order
recruitment	$F_k \exp(-\lambda_0 e^{\lambda_1(\mathrm{ST}-3)} \mathrm{RD}) \times$ seed source	light window × seed source: LANDIS-II establishment	form; fitted on sapling in-growth (FIA)
CO_2, temperature	**not in the model**	design proposes $1 + \beta \ln(\mathrm{CO}_2/350)$, β from the literature	—

Established = a published, widely used form; "ours" = a design choice of this project; "FIA" = coefficients estimated from GA+FL FIA remeasurements. References given from memory: check before presenting.

Backup: Stand delineation (GA + FL) — inputs and land status

Inputs (all on the TreeMap 30 m EPSG:5070 grid; 1 px = 0.09 ha)

- TreeMap 2022 plot ids + attributes (BA, height, QMD, cover, TPA, softwood share)
- Hansen GFC v1.12: loss year, tree cover 2000, land mask
- LANDFIRE EVT 2022: excluded types; tree / shrub / herb life form
- AlphaEarth 2022, 2025: 64-d embeddings, de-quantised $\mathrm{sign}(q)(q/127.5)^2$
- SOLUS 0–30 cm: clay, sand, SOC, AWC, pH, restrictive depth
- Barriers (1-px lines): TIGER roads and water, FL parcels

Status classes (recent = Hansen loss ≥ 2017; former forest = cover_{2000} ≥ 25%)

- **1 forest**: any TreeMap pixel
- **2 harvested**: recent loss on former forest, EVT not excluded (overrides 1)
- **3 regenerating**: not TreeMap, not excluded, and older loss, or a disturbed / plantation EVT, or shrub/herb with cover_{2000} ≥ 50%
- **4 forest unmapped**: EVT tree life form, not TreeMap
- **9 converted**: loss on former forest, now an excluded EVT (not forest land)
- 0 outside the AOI and on Hansen non-land

Speckle cleaning: same-class patches < 2 ha take the class of the nearest larger forest-land patch (≤ 15 px).

scripts 10_forest_land.py, 11_land_clean.py, config/pilot.yml

Backup: Stand delineation — the per-pixel feature vector (34 dimensions)

Each group is projected and scaled so that its total variance on a fit sample (60 tiles × 4000 forest-land pixels) equals w^2:

$\mathbf{y}_{\text{AEF}} = (\mathbf{a}_{25} - \mu)\, V_{16}^{\mathsf{T}}\ 1.0\ /\sqrt{\sum_{k\le 16}\lambda_k}$	16 PCs, AlphaEarth 2025
$\mathbf{y}_{\text{old}} = (\mathbf{a}_{22} - \mu)\, V_{8}^{\mathsf{T}}\ 0.4\ /\sqrt{\sum_{k\le 8}\lambda_k}$	AlphaEarth 2022 on the 2025 basis (8)
$y_{\text{chg}} = \big(1 - \cos(\mathbf{a}_{25}, \mathbf{a}_{22})\big)\ 0.5\ /\ \text{sd}(1-\cos)$	change since 2022 (1)
$\mathbf{y}_{\text{TM}} = (\mathbf{x} - \mu)/\sigma\ \cdot\ 0.6/\sqrt{6}$	TreeMap attributes (6)
$\mathbf{y}_{\text{soil}} = \mathbf{z}\, V_{3}^{\mathsf{T}}\ 0.3\ /\sqrt{\sum_{k\le 3}\lambda_k}$	3 PCs of the z-scored soils

- TreeMap: ln BA, height, ln QMD, cover, ln TPA, softwood share; zeroed (= sample mean) off TreeMap forest or where harvested.
- The weights are per-group standard deviations: AlphaEarth dominates, TreeMap and soils refine.

scripts/features.py (Featurizer.features), 20_fit_features.py

Backup: Stand delineation — superpixels and the region-adjacency merge

1. Superpixels — SLIC per 512-px tile and per status, on forest land minus barrier pixels: $n_{seg} = \lfloor n_{px} / 11 \rfloor$ (1 ha), connectivity enforced; never cross barriers, statuses or tiles. **Input: raw AlphaEarth 2022 ⊕ 2025**, compactness 0.15 (next slide); the merge uses the 34-d stack.

2. Merge on the region-adjacency graph (edges = shared boundary L_{ab}):

$$c_{ab} = \underbrace{\frac{n_a n_b}{n_a + n_b} \, \|\mu_a - \mu_b\|^2}_{\text{Ward}} \times s_{ab}$$

$$s_{ab} = 1 + 0.5\left(1 - \min\left(1, \frac{L_{ab}}{\min(P_a, P_b)}\right)\right)$$

(s_{ab} favours long shared boundaries; P = perimeter) same status, $n_a + n_b \le 200$ ha; each round mutually-best pairs with $c_{ab} < \tau$ merge.

3. Threshold τ — bisected (14 steps in $[10^{-2}, 10^{4}]$, 300 sample components) to an area-weighted median stand of 15 ha.

4. Minimum mapping unit — then regions < 22 px (2 ha) merge into their cheapest neighbour of the same status.

5. Finalise — barrier and sliver pixels are grown into 4-adjacent stands (same status first), so stands stay connected; fragments < MMU join the same-status neighbour with the longest shared boundary, only toward larger ones (acyclic).

Outputs — `stands.tif`, polygons: status, area, loss year, TreeMap means, main plot, soils.

scripts 21_superpixels.py, 22_merge.py, rag_merge.py, 23_finalize.py

Backup: Generative vegetation — FIA plots without their true locations

Goal: stand-level vegetation (cohorts: species, age, *N*, QMD, *H*, CR) mapped from rasters, *independent of TreeMap*, learned from FIA.

Training table (GA+FL, 8,712 plots, 34,300 visits since 2000): cohort tokens per visit (species × age class: $\ln N$, $\ln$QMD, *H*, CR) per forested area; consecutive visits = pairs.

The location is latent: public coordinates are fuzzed within ≈1.6 km. Candidates: 90 m footprints of forest land in the disk, ≤256 per plot (stratified, inclusion-weighted), with AlphaEarth 2022 / 2025, disturbance history 2000–2025 (LANDFIRE class, severity; Hansen loss), soils.

Location prior from FIA's own records: TRT / DSTRB codes matched *by type* to the footprint's evidence in the event year:

- cutting / site prep. / regeneration ↔ harvest, thinning, chemical, fire, *unknown*, Hansen loss: LR 4
- crown fire: 4; ground fire, insects, weather, human: 2
- records of one event (within 2 yr) count once
- no code over a remeasurement ⇒ stand-replacing evidence in that period ×0.25

History consistency: footprints compatible with *all* detectable records, in time order (unsatisfiable ones skipped).

Location information per plot is tiny (≈0.003 nats): the records anchor the *dynamics*, not the place.

g02b_build_gafl.py (EVK), g02e_fia_records.py, g02g_history_consistency.py; location tests g02f, g08

Backup: Generative vegetation — temporal JEPA with disturbance jumps

$$z_v = E(\text{tokens}_v) \in \mathbb{R}^{16}$$ masked set encoder, SIGReg toward $N(0, I)$

$$\frac{dz}{dt} = -\left(\frac{1}{\tau} + g\right) \odot z + g \odot A, \quad g = \sigma(W[z, s] + b)$$ liquid time-constant flow (yearly), site s

$$z \leftarrow z + J\left([z,\ \text{class},\ \text{severity},\ \text{Hansen}]\right)$$ jump at a candidate's disturbance years

$$L_{\text{MIL}} = -\log \frac{1}{K} \sum_{c=1}^{K} e^{-d_c / T}, \quad c \sim \text{prior (iid)}$$ $$d_c = \| h_{22}(z_{2022}) - a_{22}^{c} \|^2 + \| h_{25}(z_{2025}) - a_{25}^{c} \|^2,\ K{=}48$$

- **pairs**: $z(v_1)$ flowed along FIA-consistent candidate histories to $v_2 \approx z(v_2)$ and v_2's cohorts: the records anchor the jumps
- **decoder**: cohort slots (presence + $\ln N$, $\ln$QMD, H, CR); summary loss TPH / BA / QMD + Bray–Curtis
- **A**: landscape prior $p(z \mid X_L, \text{year})$, mixture-density net (8 Gaussians) on the whole 1.6 km disk — robust to the fuzzing
- **B**: likelihood $p(\text{AEF} \mid z, \text{history})$, fine-tuned on well-localised plots

scripts g03b_train_gafl.py (E, flow, jumps, heads, MIL, pairs, decoder), g04_bayes_downscale.py (A, B, calibration)

Backup: Generative vegetation — mapping stands, verification, results

Mapping (per stand of the delineation): X_L of the nearest 1 km landscape node (same descriptor as the plots' disks) → z ~ A at 2021 → optionally weighted by B on the stand's footprints → flow to 2025 through the footprints' disturbance histories → decode; calibration factors from held-out calibration plots (15 % of training blocks).

Pipeline `genveg_pipeline.py`: table → records → history → stand footprints → train (2 seeds) → dynamics test → downscale → map → Forest Explorer → verify: **45 independent checks** (SQL / direct raster reads / re-implemented specification) pass.

Held-out visits 2022–2025, mapping path (median $|\log \text{error}|$; spatial blocks, 3.2 km buffer)

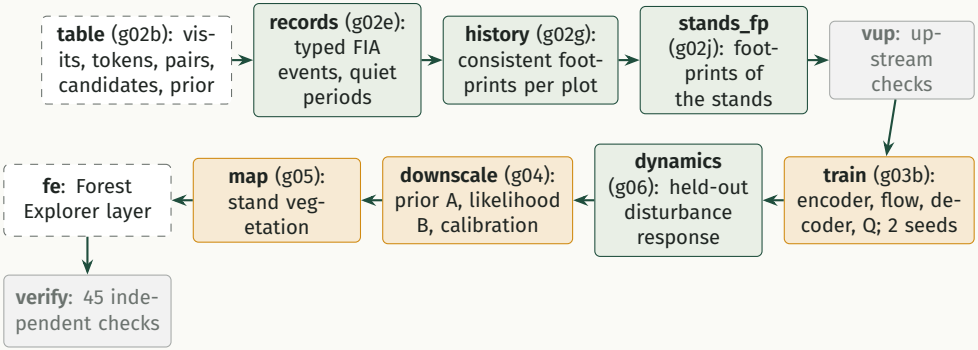
	TPH	BA	QMD
generative (calibrated)	0.611	**0.412**	**0.308**
kNN on landscape features	0.664	0.414	0.359
null (median)	**0.609**	0.453	0.327

- modest: beats kNN on QMD, null on BA; no better than null on TPH
- dominant-species accuracy 0.42 vs kNN 0.36 (majority 0.22)
- open: the direct predictor's BA level varies with the training seed (log bias −0.27 … − 0.62; calibration absorbs it)

scripts g05_map_stands.py, gv_verify.py; docs/GENVEG_PIPELINE.md, GENERATIVE_VEG_PLAN.md

Backup: generative vegetation in detail

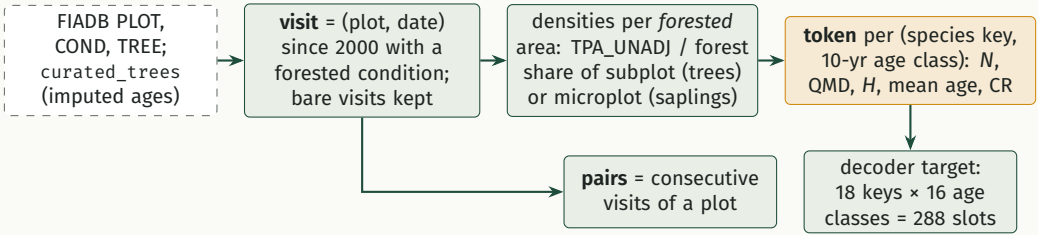
Backup: generative vegetation — the pipeline, stage by stage



- one driver `genveg_pipeline.py` + `config/genveg.yml`; each stage hashed (config, code, upstream, pixel content of the stand rasters) and skipped when current
- inputs: FIADB (read-only), `curated_trees`, AlphaEarth 2022 / 2025, LANDFIRE disturbance 2000–2025, Hansen loss year, SOLUS, the stand run's `land_clean` and `stands.tif`

`standmap/docs/GENVEG_PIPELINE.md`, `scripts/genveg_pipeline.py`

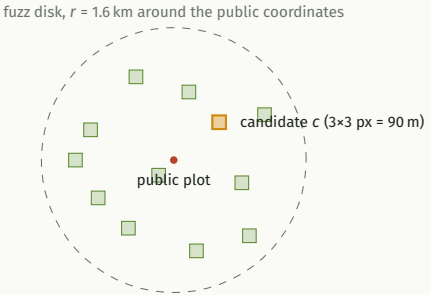
Backup: generative vegetation — step 1a, visits and cohort tokens



- species keys: the 16 commonest live species + catch-alls _H (hardwood), _S (softwood)
- age class = ⌊age/10⌋, clipped to 0–15; N = ∑ TPA·2.471/share; QMD TPA-weighted quadratic mean; H, age, CR TPA-weighted means
- table: 8,712 plots, 34,300 visits, 270,488 tokens (≤32 per visit), 25,588 pairs
- vetting: visits with forest share < 0.25 excluded from training (inflated densities)

`standmap/scripts/g02b_build_gafl.py` (visits, tokens, pairs); checks T1a–T1j; run `data/proto/gafl_pipe`

Backup: generative vegetation — step 1b, candidate footprints and their features



footprints on forest land (status 1–4), stride 3 px; ≤256 kept per plot (high-prior stratum + random fill), each with an inverse inclusion weight so disk statistics stay unbiased

footprint feature vector (394-d)

AEF 2022 64	AEF 2025 64	disturbance 26 yr × 10	soil 6

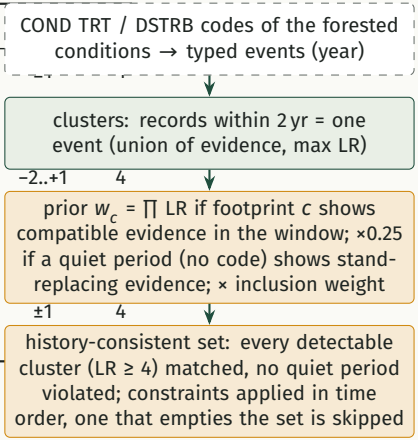
- AlphaEarth: 3×3 mean of the de-quantised 64-d embedding
- per year 2000–2025: fraction of the 9 pixels in 8 LANDFIRE classes (harvest, thin, fire, insects/disease, weather, chemical, development, unknown change) at the event's calendar year, max severity, Hansen loss fraction
- SOLUS 0–30 cm at the centre: clay, sand, SOC, AWC, pH, restrictive depth
- 2.19 M candidate rows in all

`standmap/scripts/g02b_build_gafl.py` (`feats`, `plot_job`); checks T3a–T3i, T5; run `data/proto/gafl_pipe`

Backup: generative vegetation — step 2, FIA records, location prior, consistent histories

FIA record	compatible evidence
T10 cutting	harvest, thin, unknown, Hansen
T20 site preparation	harvest, chem., fire, unknown, Hansen
T30 artificial regen.	harvest, chem., unknown, Hansen
T40 natural regen.	harvest, fire, unknown, Hansen
T50 other treatment	thin, chem.
crown fire	fire, Hansen
ground fire, insects, weather, human	own class

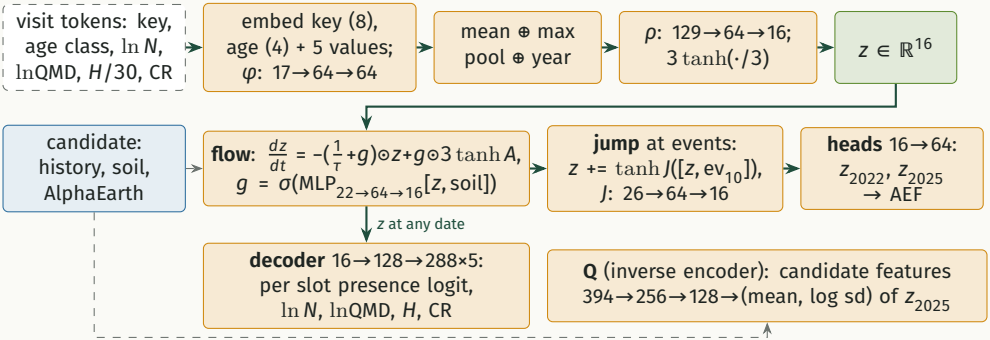
classes and lifts from the data (`g09`): e.g. cutting → harvest ×4.4–4.8, Hansen ×2.5–2.8; mild records never match Hansen



The set constrains which disturbance *histories* a plot can have, not where it is: held-out location information ≈0.003 nats per plot.

`standmap/scripts/g02b` (`EVK`, `clusters`, `prior_weights`), `g02e_fia_records.py`, `g02g_history_consistency.py`; checks T3f, T7, T10; run `data/proto/gafl_pipe`

Backup: generative vegetation — step 3, model architecture



τ per latent dimension (learned; initial $e^2 \approx 7.4$ yr); A = learned attractor. Decoder slots: 18 species keys × 16 ten-year age classes. Q is the earlier mapping model (direct predictor); the final mapping uses A × B (step 4).

`standmap/scripts/g03b_train_gafl.py` (classes `Enc`, `Flow`, `Dec`, `Q`; heads `h22`, `h25`); run `data/proto/gafl_pipe`

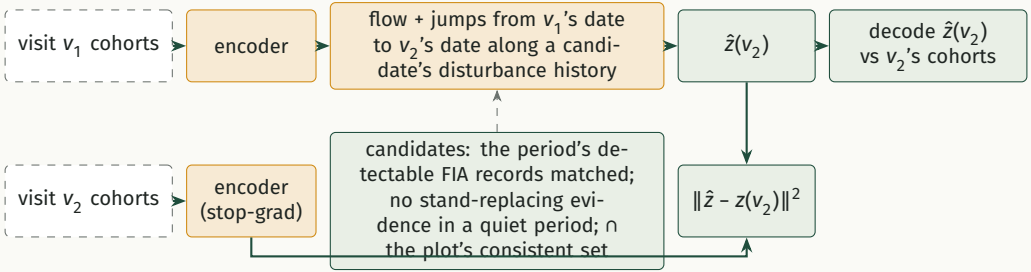
Backup: generative vegetation — step 3, losses and training

term	definition	weight
MIL	K = 48 candidates drawn iid from the (history-consistent) prior; d_c = AlphaEarth error of the flowed z at 2022 / 2025 (years after the visit); $-T \log \frac{1}{K} \sum_c e^{-d_c/T}$, T = 0.05	1
decoder	slot BCE (class-weighted) + masked MSE of values + summary error (TPH, BA, QMD, Bray–Curtis of BA shares) + absent-slot pull	1
pair	flowed $z(v_1)$ at v_2's date vs stop-grad $z(v_2)$	0.5
pair decode	flowed $z(v_1)$ decoded vs v_2's cohorts	0.5
SIGReg	Epps–Pulley test of 32 random 1-D projections of z vs $N(0,1)$ (LeJEPA), $\times\sqrt{B}$	1
Q	Gaussian NLL of z_{2025}, posterior-weighted; decoded Q mean vs the visit's cohorts	0.2, 0.5

- **split**: 20 % of 10-km blocks held out; training plots within 3.2 km of a test plot excluded (disks could overlap): 26,291 train / 6,732 test visits
- **curriculum**: first 4 of 14 epochs on the 5,729 visits of plots with ≤50 consistent footprints
- AdamW, lr 10^{-3}, weight decay 10^{-4}, batch 64, gradient clip 5; 2 seeds (seed 0 used downstream)
- the encoder sees the visit's own cohorts; satellite data enter only through the heads (MIL) and Q

`standmap/scripts/g03b_train_gafl.py` (training loop); `config/genveg.yml` `train.env`; checks T4a–T4j, T12; run `data/proto/gafl_pipe`

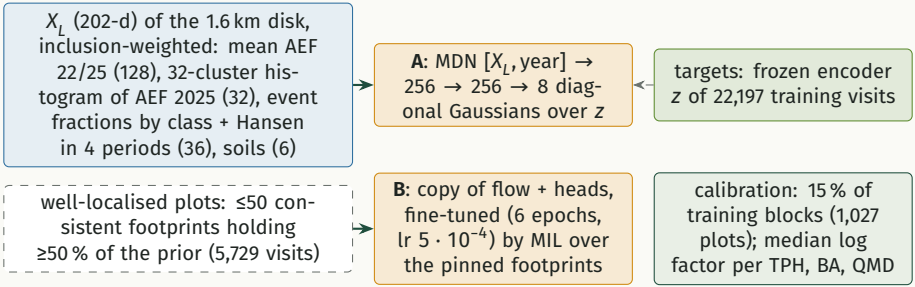
Backup: generative vegetation — step 3, FIA-anchored remeasurement pairs



- the plot's own FIA records select which LANDFIRE / Hansen histories the flow may take: the records anchor the jumps
- $\hat{z}(v_2)$ is also decoded against v_2's cohorts (a clear-cut must decode to few trees)
- training only: the measured BA ratio v_2/v_1 < 0.3 requires stand-replacing evidence, > 0.6 forbids it
- 25,213 pairs; 5,091 with a detectable event (5,065 matched); 18,235 constrained by a quiet period

`standmap/scripts/g03b_train_gafl.py` (`pair_cand`); checks T8, T8b; run `data/proto/gafl_pipe`

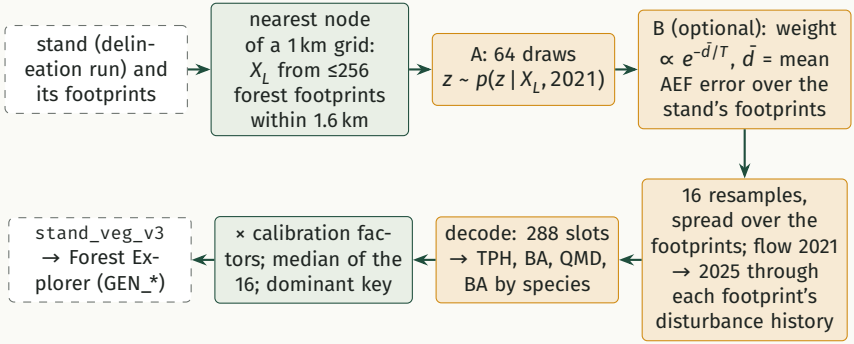
Backup: generative vegetation — step 4, landscape prior A, likelihood B, calibration



- A describes the plot as a random draw from the forest of its whole disk: robust to the coordinate fuzzing. MDN NLL train −25.5, test −23.8
- B is used only as a likelihood; the original flow stays the vegetation dynamics the decoder was trained on. Calibration log factors of the mapping path: 0.19 (TPH), 0.20 (BA), 0.06 (QMD)

`standmap/scripts/g04_bayes_downscale.py` (`landscape`, `MDN`, B fine-tuning, `map_path`); check T5; run `data/proto/gafl_pipe`

Backup: generative vegetation — step 5, mapping one stand



- stand landscapes are built exactly as the plots' disks (same footprint grid, subsample, descriptor), so A sees the same kind of input it was fitted on
- two maps: without B (`B0`) and with B (`B1`); the last one written is the default layer. The same path is validated on held-out plots (step 6)

`standmap/scripts/g05_map_stands.py`, `g02j_stand_footprints.py`; checks T6a–T6c, T11; run `data/proto/gafl_pipe`

Backup: generative vegetation — step 6, evaluation and verification

median | log error|, held-out	TPH	BA	QMD
mapping path (visits 2022–25, *n* 1,058)	0.611	**0.412**	**0.308**
kNN on X_L	0.664	0.414	0.359
null (median, ±3 yr)	**0.609**	0.453	0.327
A alone, all held-out (*n* 1,733)	0.643	0.460	0.330
A × B, localised plots (*n* 333)	0.791	0.451	0.313
A alone, same plots	0.706	0.460	0.320

Localisation by B: posterior mass on the pinned footprints 0.048 vs their share 0.049 ⇒ no location skill.
Dynamics (seed 0, held-out): own-visit reconstruction BA corr 0.96, slope 0.90; synthetic clear-cut → BA ×0.08, fire ×0.94; 996 cut pairs: log BA error 0.67 vs persistence 0.59 (change corr 0.53).

45 independent checks (`gv_verify`, `gv_verify_upstream`): SQL against FIADB, direct raster reads, re-implemented specifications.

- T1: visits, keys, tokens, pairs, BA = FIA formula
- T2–T3: plot coordinates, candidates, AlphaEarth, soils, disturbance and Hansen fractions, prior, inclusion weights
- T4: summaries, split disjointness, flow timing, MIL sampling, SIGReg
- T5–T13: landscape unbiasedness, stand footprints, localisation, pairs, history sets, FE layer, loss targets

`standmap/scripts/g04` (`bayes_eval.json`), `g06_dynamics_check.py` (`dynamics_check_*_s0.json`), `gv_verify.py`; run `data/proto/gafl_pipe`